A Project Work II Report on

REAL-TIME FOREST ENCROACHMENT MONITORING USING GIS

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE AND DATASCIENCE

by

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CERTIFICATE

Date:

This is to certify that Project report entitled Real-Time Forest Encroachment Monitoring using GIS submitted by **Ms Bolleddula Rishitha** (Roll No. 160622747016)**,** **Ms Indrakanty Sree Anushka** (RollNo. 160622747023)), student of Department of Artificial Intelligence and Data Science, Stanley college of Engineering and Technology for Women in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in **ARTIFICIAL INTELLIGENCE AND DATA SCIENCE** as specialization is a record of bonafide work carried out in IV year VIII Sem during the academic year 2025-2026.

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DECLARATION

We hereby certify that the work which is being presented in the project entitled **"REAL-TIME FOREST ENCROACHMENT MONITORING USING GIS"** submitted in the **DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE** is an authentic record of my own work carried under the supervision of **NEHA TANVEER (ASSISTANT PROFESSOR)** This study has not been submitted to any other institution or university for the award of any other degree.

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Institute Vision & Mission

Vision: Empower Women; Impact the World. Empowering girl students through professional education integrated with values and character to make an impact in the World.

Mission :

1. Providing quality engineering education for girl students to make them competent and confident to succeed in professional practice and advanced learning. Establish state-of-art- facilities and resources to facilitate world class education.

2. Integrating qualities like humanity, social values, ethics, leadership in order to encourage contribution to society.

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Vision: To be the centre of excellence in empowering girl students with quality education to make lifelong global innovators and contributors in the ever-advancing field of Artificial Intelligence and Data Science.

Mission:

1. To provide a student-centric education that aims to focus on a four-tiered strategy of Education, Research, Development and Innovation by formulating a meaningful curriculum that combines theory and practical skills.

2. To Foster industry collaborations for impactful applications, industry readiness and encourage students in research, innovations and inventions through quality internships, hackathons and other technical events.

3. To let our young minds flourish in any industry by promoting continuous learning and develop employability skills through quality training programs.

4. To produce competent and ethical engineers who will design, develop, innovate and

invent ethical AI systems, leaving a remarkable impact on the technological needs of the society and achieve self- sustainability.

Program Educational Objectives: (PEO's)

PEO1: To provide graduates with the proficiency to utilize the fundamental knowledge of basic sciences, mathematics, artificial intelligence, data science and statistics to build systems that require management and analysis of large volume of data.

PEO2: To enrich graduates with necessary technical skills to pursue pioneering research in the field of Artificial Intelligence

PEO3: To encourage students to think critically, develop innovative skills, expose them to an array of ideas and information through numerous technical events, hackathons and quality internships.

Program Outcomes (AY:2024-2025)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to vnational & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

Program Specific Outcomes: (PSO's)

PSO1: To instill interest and curiosity in students in the field of Artificial Intelligence and Data Science through project-based learning.

PSO2: To provide a concrete foundation and enrich their abilities to qualify for employment, higher studies and pursue research in Artificial Intelligence and Data Science with ethical values.

PSO3: To promote ethical and responsible AI practices for the benefit of humanity and to harness Artificial Intelligence for a positive societal impact & meet global standard.

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Course Objectives:
1. To enhance practical and professional skills
2. To familiarize tools and techniques of systematic Literature survey and documentation.
3. To expose the students to industry practices and team work.
4. To encourage students to work with innovative and entrepreneurial ideas.

Course Outcomes: The course should enable the students to:
1. Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to real-world problems.
2. Evaluate different solutions based on economic and technical feasibility.
3. Effectively plan a project and confidently perform all aspects of project management.
4. Demonstrate effective written and oral communication skills.
5. Analyze, apply and appreciate contemporary project management tools and methodologies in Indian context

COPO Mapping:

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PSO 1	PSO 2	PSO 3
SPW811AD.1	3	2	1	1	1	1	1	1	1	2	1	3	2	1
SPW811AD.2	3	3	2	2	2	1	1	1	1	2	1	3	3	2
SPW811AD.3	2	3	3	3	3	1	1	1	1	2	2	3	3	3
SPW811AD.4	2	2	3	3	3	2	1	1	1	3	2	3	3	3
SPW811AD.5	1	1	1	2	3	2	1	1	2	3	3	2	3	3

Knowledge and Attitude Profile (WK)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

WK → PO Mapping

WK	Mapped POs
WK1	PO1
WK2	PO1, PO2, PO4
WK3	PO1, PO2, PO3
WK4	PO2, PO3, PO4
WK5	PO3, PO7
WK6	PO5, PO11
WK7	PO6, PO7, PO10, PO11
WK8	PO4, PO12
WK9	PO8, PO9, PO12

Justification for WK → PO Mapping

WK1 → PO1

WK1 provides the scientific foundation required for application of engineering knowledge in PO1.

WK2 → PO1, PO2, PO4

Mathematical modelling and computational tools (WK2) are essential for solving problems (PO1), analyzing systems (PO2), and conducting investigations (PO4). **WK3 → PO1, PO2, PO3**

Engineering fundamentals (WK3) enable knowledge application (PO1), structured problem analysis (PO2), and systematic design of solutions (PO3).

WK4 → PO2, PO3, PO4

Specialist knowledge (WK4) supports advanced problem-solving (PO2), complex system design (PO3), and technical investigations (PO4).

WK5 → PO3, PO7

Design considerations involving sustainability and lifecycle thinking (WK5) align with solution development (PO3) and environmental responsibility (PO7).

WK6 → PO5, PO11

Engineering practice knowledge (WK6) underpins modern toolusage (PO5) and project management applications (PO11).

WK7 → PO6, PO7, PO10, PO11

Societal awareness and responsibility (WK7) relate to societal engineering roles (PO6), sustainability (PO7), professional communication (PO10), and management decisions (PO11).

WK8 → PO4, PO12

Research engagement and critical thinking (WK8) are required for investigations (PO4) and life-long learning (PO12).

WK9 → PO8, PO9, PO12

Ethics and inclusive professional conduct (WK9) align with ethical practice (PO8), team work (PO9), and professional growth (PO12).

List of all 17 United Nations Sustainable Development Goals (SDGs):

SDG No.	Goal Title
SDG 1	No Poverty
SDG 2	Zero Hunger
SDG 3	Good Health and Well-being
SDG 4	Quality Education
SDG 5	Gender Equality
SDG 6	Clean Water and Sanitation
SDG 7	Affordable and Clean Energy
SDG 8	Decent Work and Economic Growth
SDG 9	Industry, Innovation and Infrastructure
SDG 10	Reduced Inequalities
SDG 11	Sustainable Cities and Communities
SDG 12	Responsible Consumption and Production
SDG 13	Climate Action
SDG 14	Life Below Water
SDG 15	Life on Land
SDG 16	Peace, Justice and Strong Institutions
SDG 17	Partnerships for the Goals

This Project "Project Title: Real Time Forest Encroachment Monitoring Using GIS" is mapping the following SDGs:

SDG No.	SDG Title	Relevance to Project
SDG13	Climate Action	The project helps in monitoring illegal encroachment and deforestation, reducing carbon emissions and supporting climate resilience.
SDG15	Life on Land	Ensures protection of forests by identifying encroachment in real time, aiding conservation and biodiversity protection.
SDG11	Sustainable Cities and Communities	Supports sustainable land-use planning by preventing unplanned expansion into forest areas.
SDG9	Industry, Innovation and Infrastructure	Uses advanced GIS, AI, and remote sensing technologies to create innovative monitoring infrastructure

PO → SDG Mapping

PO	Mapped SDGs
PO1	SDG4, SDG9
PO2	SDG4, SDG9
PO3	SDG9, SDG12, SDG13
PO4	SDG4, SDG9
PO5	SDG9
PO6	SDG11, SDG16
PO7	SDG7, SDG11, SDG12, SDG13
PO8	SDG16
PO9	SDG5, SDG10, SDG16
PO10	SDG16, SDG17
PO11	SDG8, SDG9
PO12	SDG4, SDG9

Justification for PO → SDG Mapping

PO1 → SDG4, SDG9

Engineering knowledge contributes to qualityeducation (SDG4) and supports innovation and infrastructure development (SDG9).

PO2 → SDG4, SDG9

Problem-solving and analytical skills enhance technical education (SDG4) and foster innovative industrial solutions (SDG9).

PO3 → SDG9, SDG12, SDG13

Design of engineering solutions promotes industrial innovation (SDG9), responsible pro duction (SDG12), and climate-conscious engineering (SDG13).

PO4 → SDG4, SDG9

Research and investigations strengthen higher education quality (SDG4) and technolog ical advancement (SDG9).

PO5 → SDG9

Modern engineering tools directly support technological innovation and infrastructure (SDG9).

PO6 → SDG11, SDG16

Engineering solutions addressing societal and safety concerns contribute to sustainable communities (SDG11) and institutional integrity (SDG16).

PO7 → SDG7, SDG11, SDG12, SDG13

Environmental responsibility aligns with clean energy (SDG7), sustainable cities (SDG11), responsible consumption (SDG12), and climate action (SDG13).

PO8 → SDG16

Professional ethics supports justice, accountability, and strong institutions (SDG16).

PO9 → SDG5, SDG10, SDG16

Teamwork and inclusiveness promote gender equality (SDG5), reduced inequalities (SDG10), and institutional responsibility (SDG16).

PO10 → SDG16, SDG17

Effective communication strengthens institutional transparency (SDG16) and partner ships (SDG17).

PO11 → SDG8, SDG9

Project management skills enhance economic growth (SDG8) and industrial innovation (SDG9).

PO12 → SDG4, SDG9

Life-long learning sustains quality education (SDG4) and continuous technological inno vation (SDG9).

ABSTRACT

As the global community faces escalating threats from climate change and illegal deforestation, the need for automated, high-precision ecological monitoring has never been more critical. This seminar proposes an advanced Forest Image Classification System that utilizes a sophisticated hybrid architecture—integrating **CNNs** for spatial feature extraction, **RNN-LSTMs** for tracking temporal changes, and **Vision Transformers (ViTs)** for global contextual awareness—to accurately identify forest types and degradation patterns in real-time. By further employing **Generative Adversarial Networks (GANs)** to overcome the hurdle of limited labeled data through synthetic image generation, the system provides a scalable and robust solution for forest conservation, disaster response, and sustainable land management.

Key Words : Forest image classification, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Vision Transformers (ViTs), Generative Adversarial Networks (GANs)

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

AI	Artificial Intelligence
ReLU	Rectified Linear Unit
FPS	Frames Per Second
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
ViT	Vision Transformer
GAN	Generative Adversarial Network
GIS	Geographic Information System

CHAPTER 1

INTRODUCTION

Forests are among the most essential ecological assets on Earth, functioning as natural regulators of the climate, protectors of biodiversity, and providers of numerous environmental services that sustain human life. They contribute significantly to carbon sequestration, continuous oxygen supply, soil stabilization, and hydrological cycle regulation. Additionally, forests serve as habitats for countless plant and animal species, making them indispensable for maintaining ecological balance. Despite their unparalleled importance, forests around the world are increasingly threatened by anthropogenic activities and natural disasters. Rapid deforestation driven by expanding agriculture, urbanization, illegal logging, and mining has caused substantial shrinkage of forest cover in numerous regions. Moreover, recurring forest fires—both natural and human-induced—have intensified ecological vulnerability, leading to habitat destruction, decreased air quality, and long-term soil degradation. Illegal encroachments into protected forest zones further accelerate biodiversity loss and hinder conservation efforts. These challenges highlight the urgent need for intelligent, automated systems capable of continuously monitoring forest landscapes and identifying potential threats in a timely and efficient manner.

In recent years, significant advances in remote sensing technologies have opened new possibilities for large-scale environmental monitoring. Satellite imagery, aerial surveys, and drone-based imaging systems have made it possible to capture high-resolution forest data across vast terrains at frequent intervals. These technologies provide valuable insights into vegetation health, forest density, canopy patterns, fire outbreaks, and human activities within forest regions. However, the massive volume, complexity, and heterogeneity of this visual data demand advanced computational

The proposed project adopts such a hybrid architecture by integrating Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory Networks (LSTMs) into a unified framework for real-time forest image classification. Each component of this hybrid model plays a critical role in addressing the challenges associated with the analysis of large-scale forest imagery captured from satellites and drones. The CNN component serves as the foundation for spatial feature extraction. CNNs are specifically designed to identify patterns such as textures, edges, shapes, and spatial variations, which are common in high-resolution forest images. Whether distinguishing between forest and non-forest zones, detecting burnt areas, or recognizing degraded vegetation patches, CNNs provide highly effective feature maps that capture the spatial richness of the imagery.

While CNNs excel in spatial analysis, they fall short in modeling temporal relationships—changes that occur across sequences of images taken over time. Forest landscapes are inherently dynamic, with variations occurring due to seasonal differences, natural growth cycles, or the progression of forest fires and encroachments. To capture such temporal dependencies, the framework incorporates RNNs and LSTMs. These architectures are designed to process sequential data, making them ideal for analyzing time-series imagery or video sequences captured from surveillance drones. LSTMs, with their enhanced memory capabilities, effectively handle long-term dependencies and prevent information loss during sequence processing. This enables the system to detect slow-developing forest degradation patterns or progressive fire spread that might not be evident from a single image..

The integration of CNN, RNN-LSTM,-based techniques results in a powerful and comprehensive framework capable of real-time forest monitoring and classification. The real-time capability is crucial because early detection of forest fires, illegal encroachments, or rapid deforestation can significantly reduce damage and allow for swift intervention. When implemented with cloud platforms, edge devices, or integrated into drone systems, this model can continuously monitor forest landscapes, automatically categorize regions, detect anomalies, and generate alerts.

Figure 1.1 Non fire vs Fire

Figure 1.2 Wild fire image

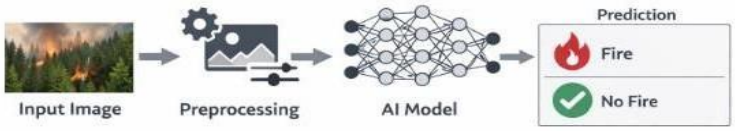

Figure 1.3 Flow chart

CHAPTER 2

LITERATURE SURVEY

"Forest fire surveillance systems: A review of deep learning methods," provides a comprehensive review of the current state-of-the-art in forest fire detection and surveillance using deep learning (DL) techniques. Given the increasing threat of forest fires due to climate change, the review systematically analyzes how DL models are applied across five main areas: classification, detection, combined detection and classification, segmentation, and combined segmentation and classification. For each application, the authors compiled and assessed details regarding the datasets utilized, data augmentation strategies, and the specific deep model architectures employed in various studies. The core goal is to summarize the effectiveness and methodologies of DL approaches to provide timely and accurate fire monitoring, ultimately advancing the technology available for mitigating the severe environmental, economic, and social consequences of forest fires.[1]

Global change and fire ecology: an overview," reviews how **global change factors**—primarily **climate change**, but also shifts in **land use** and **species distribution**—are profoundly affecting the **dynamics of fire regimes** across the planet. The authors stress that these changes are altering the fundamental relationship between fire, ecosystems, and human societies, driving both an increase in the **frequency and intensity of large, destructive wildfires** in many regions and, conversely, a **decline in fire activity** in others due to ecosystem transformations and human intervention. The review emphasizes the complex, non-linear interactions between these global change drivers, which makes fire management and prediction challenging, and calls for **integrated, proactive strategies** that consider these multifaceted impacts to effectively manage future fire risk and promote ecosystem resilience.[2]

"A Low-Cost UAV-Based Fire Detection System Using Thermal and Visible Cameras," presents an efficient and cost-effective **Unmanned Aerial Vehicle (UAV)** system designed for **early-stage fire detection** in forests and difficult-to-access areas. The system strategically utilizes a combination of both **thermal and visible-light cameras** to enhance detection accuracy and reduce false alarms, where the thermal camera is critical for detecting the heat signature of incipient fires, and the visible-light camera confirms the presence of flame or smoke. By leveraging the UAV's superior mobility and overhead vantage point, the system is capable of **real-time image acquisition and processing** through embedded algorithms used to precisely pinpoint potential fire sources based on analyzed heat signatures and visual cues. This integration of readily available, low-cost hardware and specialized, multi-sensor fusion algorithms aims to provide a **reliable, timely, and scalable** solution for expansive forest fire surveillance, effectively surpassing the common limitations of traditional fixed-sensor networks or less-responsive satellite-based monitoring methods.[3]
"Review of Modern Forest Fire Detection Techniques: Innovations in Image Processing and Deep Learning," comprehensively reviews the evolution of forest fire surveillance, emphasizing the move from limited **traditional sensor-based and satellite methods**—which suffer from slow response times and small coverage—to sophisticated **image processing (IP)** and **deep learning (DL)** techniques. While early IP methods analyzed visual cues like color, motion, and texture, they were prone to high rates of false alarms from non-fire objects. The review highlights **Deep Learning** as the key innovation, showcasing how models like **Convolutional Neural Networks (CNNs)** for classification and **YOLO or Faster R-CNN** for object detection and segmentation now enable highly accurate, real-time fire and smoke identification by learning complex feature patterns directly from imagery. Ultimately, the paper concludes that the integration of these DL models, particularly on **Unmanned Aerial Vehicles (UAVs)**, represents the most robust and promising direction for achieving scalable, low-latency, and reliable forest fire surveillance.[4]
"Impact of climate change and land use/land cover changes on wildfire risk in a Mediterranean region: a systematic review and meta-analysis," synthesizes existing

research to comprehensively evaluate how the intertwined factors of **climate change** and **land use/land cover (LULC) changes** collectively influence wildfire risk, specifically within vulnerable **Mediterranean regions**. The systematic review confirms that rising temperatures, altered precipitation patterns, and increased drought frequency due to climate change are driving a significant escalation in fire danger by creating more favorable conditions for ignition and rapid spread. Furthermore, the analysis emphasizes that concurrent LULC changes—such as **rural abandonment** leading to fuel accumulation and landscape fragmentation caused by **urban expansion**—critically modify ecosystem flammability and fire behavior, interacting synergistically with climatic changes to compound the overall wildfire hazard. The findings underscore the necessity for integrated fire management and land planning strategies that account for these combined, dynamic environmental pressures to effectively mitigate future wildfire risks in Mediterranean landscapes.[5]

"Review of Modern Forest Fire Detection Techniques: Innovations in Image Processing and Deep Learning," conducts a systematic review of the shift in forest fire surveillance from limited **traditional sensor-based and satellite methods**—which are often slow, require proximity, or lack the necessary spatial and temporal resolution for early detection—to highly effective **vision-based systems** powered by advanced computing. It highlights that while early **Image Processing (IP)** techniques used low-level features like color thresholds, motion, and texture to identify fire and smoke, they suffered from frequent **false alarms** due to confounding environmental factors. The review then establishes **Deep Learning (DL)** as the current technological frontier, showcasing its superior performance through the use of **Convolutional Neural Networks (CNNs)** for classification and sophisticated frameworks like **YOLO (You Only Look Once)** and **Faster R-CNN** for **real-time object detection and segmentation**. These models excel by automatically learning hierarchical and abstract features, enabling them to precisely locate fire and smoke plumes under varying conditions. A critical finding is the increasing integration of these powerful DL models with **Unmanned Aerial Vehicles (UAVs)**, which provide mobile, high-resolution aerial vantage points, making the combined system the most promising and reliable

direction for achieving scalable, low-latency, and accurate early-stage wildfire detection and mitigation.[6]

"Forest Fire Image Classification via Hybrid Deep Learning and Stacking Ensemble Technique," addresses the crucial need for highly accurate and timely fire detection by proposing a novel **hybrid model** that integrates the strengths of **Deep Learning (DL) feature extraction** with **Machine Learning (ML) ensemble classification**. The core methodology involves using pre-trained DL models, specifically **InceptionV3**, **SqueezeNet**, and **DeepLoc**, as **feature extractors** to generate robust representations (embeddings) from forest images, which are then fed into multiple diverse classifiers, including **Neural Networks (NN), Decision Trees (DT), and XGBoost**. To further optimize predictive reliability and overcome the limitations of any single model, the research employs a **stacking ensemble technique**, where the predictions of the base classifiers are combined by a **meta-learner** (like Logistic Regression) to generate the final, most reliable classification decision. Through rigorous testing, the integration of **InceptionV3** with NN and XGBoost, orchestrated by a Logistic Regression meta-learner, achieved the **highest performance metrics**, demonstrating an exceptional **98.9% accuracy** and 99.9% AUC, thereby validating the approach as a highly efficient and dependable solution for building robust early warning systems for wildfire management.[7]

"Fire modeling under global change: Challenges, opportunities, and future directions," reviews the current state of **wildfire modeling**, emphasizing the significant challenges and emerging opportunities presented by **global change** factors like climate change and altered land use. The authors underscore that predicting future fire regimes is complex because these drivers interact non-linearly to affect fuel availability, ignition probability, and fire behavior, leading to **high uncertainty** in model projections, especially regarding the potential for extreme fire events. To address this, the review advocates for a shift towards more **integrated, sophisticated modeling frameworks** that incorporate dynamic vegetation processes, complex human-fire interactions (such as management and suppression), and improved spatial and temporal resolution in

projections. It highlights the potential of **machine learning** and **process-based models** that explicitly account for model uncertainty as crucial future directions for providing reliable, actionable information to fire managers and policymakers facing increasingly volatile fire seasons.[8]

"Forest Fire Smoke Detection Using Deep Learning Convolutional Neural Network (CNN) Approach," focuses on developing an effective method for **early forest fire detection** by accurately identifying the presence of **smoke** using a **Convolutional Neural Network (CNN)**. Recognizing that smoke often appears before visible flames and is a crucial early indicator, the study proposes a custom-designed CNN model specifically trained to classify images as containing either smoke or no smoke. The researchers meticulously collected and processed a large dataset of forest images containing smoke under various conditions to train their model, aiming to overcome the limitations of traditional image processing techniques that struggle with varying smoke densities, colors, and environmental factors. The implemented CNN architecture demonstrated **high accuracy and robust performance** in distinguishing smoke from visually similar elements like clouds, thereby validating the approach as a viable and reliable component for creating automated, real-time surveillance systems essential for the prompt containment of wildfires.[9]

"A Spatio-Temporal Prediction Model Based on Machine Learning for Satellite-Based Wildfire Detection," addresses the challenge of timely and accurate wildfire detection using **satellite remote sensing data** by developing a novel **spatio-temporal prediction model** based on **Machine Learning (ML)**. Recognizing the limitations of traditional fixed-threshold methods, which are prone to false alarms and delayed detection, the research proposes utilizing **Random Forest (RF)** and **Gradient Boosting Decision Tree (GBDT)** algorithms to analyze a rich set of spectral, spatial, and temporal features derived from multi-sensor satellite imagery (e.g., MODIS and Landsat). The key innovation lies in incorporating **temporal context** by training the model to predict the presence of a fire in a given area based not only on the current image data but also on data from previous time steps, thus improving the ability to

distinguish actual hot spots from transient anomalies and geological features. The developed ML-based models demonstrate **superior performance** compared to conventional methods, significantly **reducing false alarms** and enhancing the overall **detection accuracy and timeliness**, offering a more robust and scalable framework for continuous, large-scale wildfire monitoring.[10]

"A new forest fire detection method based on vision and spatio-temporal modeling," introduces a novel, vision-based approach to enhance the accuracy and speed of **early-stage forest fire detection** by integrating advanced **image processing** with **spatio-temporal analysis**. Recognizing that traditional methods struggle with high false alarm rates and latency, the proposed system first employs robust image analysis techniques, often involving **deep learning or optimized color/motion segmentation**, to quickly identify potential fire or smoke regions within video frames captured by surveillance cameras. Crucially, the system then applies a **spatio-temporal model** which doesn't just analyze individual frames, but examines the **propagation and continuity** of the fire characteristics (like shape, movement, and temperature features) across a sequence of frames and across space. This spatio-temporal consistency check is key to **distinguishing real, developing fires from transient visual noise** or similar-looking objects (like moving clouds or sunsets), thereby significantly reducing false alarms and providing highly reliable, near real-time alerts for prompt wildfire containment efforts.[11]

The system's core is an **IoT device** utilizing a Node MCU and sensors for heat, smoke, and flame, which not only provides an immediate local alert via an Arduino buzzer but also wirelessly transmits the fire's precise location and status to a central unit, enabling automatic alerts (text/call) to forest authorities and activating an integrated **prevention system** to contain the initial spread. Complementing this rapid response is a **Machine Learning** component, where the **Random Forest** algorithm achieved a 99% accuracy in predicting fire risk based on environmental factors, allowing for proactive intervention and resource allocation in high-risk zones, thereby offering a complete and intelligent solution for forest fire management.[12]

A highly efficient, cost-effective method for forest fire management is presented, leveraging **Unmanned Aerial Vehicles (UAVs)**, or drones, equipped with cutting-edge deep learning technology. The central innovation is a **mixed learning technique** that combines the **YOLOv4 tiny** object detection model with **LiDAR** technology, all running on the UAV's onboard CPU. This setup allows the drone to conduct autonomous patrols, enabling rapid, real-time detection of fire and smoke. The system demonstrated strong performance, achieving a high classification **accuracy of 91%** (with comparison metrics up to 93.3%) and a fast processing time of only 1.24 seconds, thereby surpassing traditional fire detection techniques. A key feature is the ability to generate **3D models** of the affected forest region, providing ground stations with critical data for resource planning and targeted response. The authors acknowledge that the system's sensitivity to high smoke zones is compromised in areas with **dense fog or clouds**, which may be misclassified as fire smoke.[13]

A novel method is proposed for **Forest fire flame and smoke detection from UAV-captured images using fire-specific color features and multi-color space local binary pattern**. This approach addresses the increasing frequency and severity of forest fires by integrating computer vision with low-cost, small-sized **UAVs**. The core technique uses a **single Artificial Neural Network (ANN)** trained on challenging aerial images that feature tiny flame pixels, variable illumination, and complex, occluded backgrounds. The method employs a combination of **fire-specific color features** and a **multi-color space local binary pattern** to robustly identify both flame and smoke signatures . The system demonstrated effective performance with **F1 scores of 0.84 for flame and 0.90 for smoke**, maintaining a respectable processing speed of **19 frames per second**. Crucially, this technique **outperformed several conventional algorithms** (including Support Vector Machine, Random Forest, Bayesian Classifiers, and YOLOv3), proving its capability to accurately detect fires across a wide range of challenging aerial circumstances, making it a promising and practical solution for continuous, reliable monitoring.[14]

The latest study presents a comprehensive, deep learning-based surveillance system for the early detection and monitoring of forest fires using a custom-designed **Unmanned Aerial Vehicle (UAV)**. This system is engineered for autonomous operation, utilizing a four-rotor UAV equipped with a camera and an **NVIDIA Jetson Nano** embedded AI computer for real-time onboard data processing. The researchers compared the performance of several deep learning models for detection and classification. Specifically, they tested **YOLOv8** and **YOLOv5** for object detection, which pinpoints the exact location of the fire, and a custom **CNN-RCNN network** for image classification, which determines if a fire is present. The system achieved high accuracy across the board: **96% accuracy** for both the YOLOv8 classification and the CNN-RCNN classification, along with **89% accuracy** for both YOLOv8n and YOLOv5n object detection. The UAV transmits the detected fire images and their precise GPS coordinate information to a specialized **ground station interface**, enabling immediate, targeted intervention by civil defense and forest monitoring centers. This integration of edge computing on the drone with a ground station interface represents a proactive and highly accurate solution for overcoming the limitations of traditional fire detection methods.[15]

This study introduces a sophisticated system designed for **early forest fire detection (FFD)**, utilizing a **Universal Trust Model (UTM)** integrated within an **intelligent Wireless Sensor Network (WSN)** and enhanced by machine learning. The system is designed to significantly reduce detection time and boost reliability by partitioning the forest into clusters and intelligently positioning sensor nodes to guarantee extensive coverage. The core of the UTM is its ability to compute and assess trust ratings for every sensor node based on factors like communication stability, energy levels, and data integrity, which allows it to isolate faulty sensors and ensure the reliability of the delivered information. Furthermore, a machine learning **regression model** is deployed at the base station to augment detection precision by analyzing key environmental attributes like temperature, humidity, and CO2 concentrations. Experimental results confirm that this trust-driven approach achieves a high rate of data processing and a

reduced time delay compared to existing systems, making it a highly promising solution for proactive fire prevention and management.[16]

Real-time fire and smoke detection system tailored for smart forests, leveraging **improved deep learning models** analyzed using drone footage. The study focuses on enhancing the accuracy and speed of aerial surveillance by developing and evaluating modifications to established deep learning architectures, demonstrating their effectiveness in complex, real-world forest environments. The goal is to overcome common challenges such as minuscule fire pixels, poor illumination, and smoke blending into the background, which compromise the performance of standard vision algorithms. By deploying these optimized models on Unmanned Aerial Vehicles (UAVs), the system can capture high-resolution images and provide immediate analysis of fire and smoke signatures, facilitating a rapid and accurate response to emerging threats. The final model is proven to offer superior performance in terms of both detection accuracy and processing speed, making it a viable solution for continuous, autonomous forest fire monitoring.[17]

A novel approach to wildland fire detection addresses the critical challenge of limited annotated data by introducing a new dataset named **SWIFT**, which features a comprehensive variety of wildland fire scenarios collected from multiple viewpoints, including both ground and aerial perspectives. To leverage this data for high-precision detection, the study proposes **CT-Fire**, a **hybrid deep learning model** that combines the strengths of Convolutional Neural Networks (CNNs) and **Transformers**. This model, along with a counterpart named DC-Fire, was rigorously tested using both **RGB (visual)** and **infrared (thermal)** aerial images, demonstrating superior performance compared to existing baseline methods. Specifically, CT-Fire achieves higher detection and recognition accuracy, particularly in complex backgrounds and when identifying **small wildfire and smoke areas**. Furthermore, the model's **faster processing speed** enables its practical application for real-time, early fire detection and monitoring in both day and night conditions.[18]

Evaluation of various **deep learning models** specifically designed for real-time fire classification and detection. The core of the study is to assess the **performance, speed, and accuracy** of different convolutional neural network (CNN) architectures when identifying and localizing fire and smoke signatures in complex visual data. The goal is to determine the most effective model for rapid deployment in monitoring systems, such as those used in smart forests. The findings highlight the critical trade-offs between a model's computational demands and its overall precision, offering essential insights for selecting an optimal, high-efficiency architecture suitable for implementation on embedded systems or aerial platforms for immediate and reliable early fire warnings.[19]

Real-time forest fire detection specifically designed for **resource-constrained devices** like Unmanned Aerial Vehicles (UAVs) and ground cameras. The methodology centers on **hierarchical multi-task knowledge distillation (MTKD)**, which transfers learned information from a robust, high-performance teacher model (DenseNet201) to a smaller, more efficient student model. This process ensures that the lightweight student model retains high accuracy while requiring significantly fewer computational resources, making it ideal for edge computing. By using a specially structured wildfire dataset that organizes fire indicators hierarchically, the system is optimized to rapidly and accurately detect early-stage fires in dynamic environments. This makes the system highly adaptable and scalable for large-scale forest monitoring, addressing the critical need for faster, more reliable detection than traditional surveillance methods can provide.[20]

Paper Title	Authors	Journal/Year	Methodology	Key Results	Research Gap / Problem
A lightweight CNN model for UAV-based image classification	Deng et al.	Soft Computing (2025)	MobileNetV2 Attention and Ensemble Learning.	Lightweight for edge devices; strong generalization. 84% accuracy.	Need for lightweight models for UAVs with limited resources.
Active wildfire detection via satellite imagery and ML	Singh et al.	Natural Hazards (2025)	SVMs using spectral bands (Landsat-8).	Enhanced detection of fires and smoke. 90% accuracy.	Improving detection efficiency and accuracy using satellite data.
Forest fire and smoke detection using DL-based LwF	Sathishkumar et al.	Fire Ecology (2023)	Transfer Learning (Xception, VGG16) with LwF.	91.41% accuracyon new data; prevented knowledge loss on old data.	Overcoming catastrophic forgetting in fine-tuned models.
SWVR: Lightweight DL Algorithm for Forest Fire Detection	Jin et al.	Forests (2024)	SWVR (RepViT + SimAM) and GSConv.	79.6% accuracy; 11.8% fewer parameters; 36.5% less cost.	Balancing high accuracy with lightweight design for real-time systems.
Review of Modern Forest Fire Detection Techniques	Özel et al.	Information (2024)	Review of Image Processing, CV, and DL systems.	DL/CV enables exceptional accuracyand real-time	Limitations of traditional sensors(low accuracy)

Table 2.1 Literature Survey

CHAPTER 3

PROPOSED SYSTEM

The proposed Forest Image Classification System uses deep learning techniques to automatically classify forest images into fire and non-fire categories with high accuracy. The system starts by collecting a diverse set of forest images from sources such as satellite feeds, drones, and ground-based cameras. These images undergo preprocessing steps including resizing, normalization, and data augmentation to improve model performance and robustness.

To address class imbalance, where fire images are less frequent than non-fire images, the system applies data augmentation techniques such as rotation, flipping, and brightness adjustments to increase the diversity of fire samples. The system uses a hybrid deep learning approach combining Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) with Long Short-Term Memory (LSTM). The CNN model extracts spatial features such as smoke patterns, flame textures, and environmental characteristics from images. The LSTM model is used to analyze temporal patterns in sequences of images or video frames, helping in identifying gradual changes like smoke spread or fire growth.

Requirements

- **High-Performance Training Hardware:** A GPU-enabled system (such as NVIDIA RTX series) is required to efficiently train CNN and LSTM models on large image datasets.
- **Deep Learning Frameworks:** Software tools like PyTorch or TensorFlow are required to build and train the CNN and LSTM models, along with OpenCV for image preprocessing.
- **Data Augmentation & Management:** A proper data pipeline is required to handle large-scale image datasets and perform augmentation techniques to handle class imbalance effectively.

CHAPTER 4

DESIGN

Project Overview & Goals

The primary objective of this project is to design and develop a robust and highly accurate binary image classification system capable of distinguishing fire from non-fire scenes in forest environments. Forest fire detection is a critical environmental and safety challenge because manual observation is slow, error-prone, and difficult to deploy across large landscapes. By leveraging modern deep learning pipelines, the project aims to automate fire detection using image frames captured from satellites, drones, CCTV systems, and IoT-enabled forest cameras. The system is designed for real-time operation, meaning that the model must maintain acceptable frames-per-second (FPS) performance without compromising accuracy. The goal is to significantly reduce false positives, which can trigger unnecessary interventions, as well as false negatives, which can result in catastrophic environmental damage.

Data Sources

The dataset is curated from public forest fire datasets, drone captures, and IoT-based surveillance cameras, providing images under varied environmental, seasonal, and climatic conditions. This diversity ensures the model generalises well and covers both typical and rare fire scenarios

• Seasonal variation: dry season, monsoon, winter

Dry, monsoon, and winter images account for changes in vegetation colour, humidity, and atmospheric properties that significantly affect fire appearance.

Labels & Metadata

Each image is annotated with one of two binary labels—**Fire** or **Non-fire**—which represent the core classification task of the model. Binary labeling simplifies the training process while maintaining high interpretability in real-world applications. Correct labeling is crucial because any mislabeling directly impacts training quality and model reliability. The labeling process includes manual verification to eliminate errors from automated labeling tools. High-quality labeled data ensures stable training, reduces confusion during inference, and enhances long-term model robustness. This strong labeling framework forms the foundation of accurate model learning.

Data Quality Checks

Rigorous quality checks ensure only reliable images contribute to training. Duplicates are removed using MD5 or perceptual hashing, corrupted files are filtered out, and blurred or artifact-heavy frames are eliminated. Manual inspection corrects mislabels, as incorrectly labeled fire images can severely misguide model learning. These measures ensure stable convergence, better generalisation, and more trustworthy predictions.

Dataset Splits

The dataset is divided into training, validation, and test sets using a stratified approach to preserve the fire–non-fire class ratio across all splits. Training data optimises model parameters, validation data tunes hyperparameters, and test data provides unbiased performance evaluation. For video-based datasets, splitting is performed at the sequence level to prevent data leakage between frames of the same video. A separate holdout dataset of real-world videos is reserved exclusively for final testing under operational conditions.

Handling Class Imbalance

Class imbalance is a significant challenge as non-fire images are naturally more abundant. To address this, oversampling increases fire image representation, while class weights in the loss function penalise fire misclassifications more heavily. Targeted augmentations further expand rare fire conditions such as early-stage flames and distant smoke plumes. These combined strategies reduce model bias toward non-fire predictions and improve overall detection sensitivity without compromising precision.

4.1 Architecture

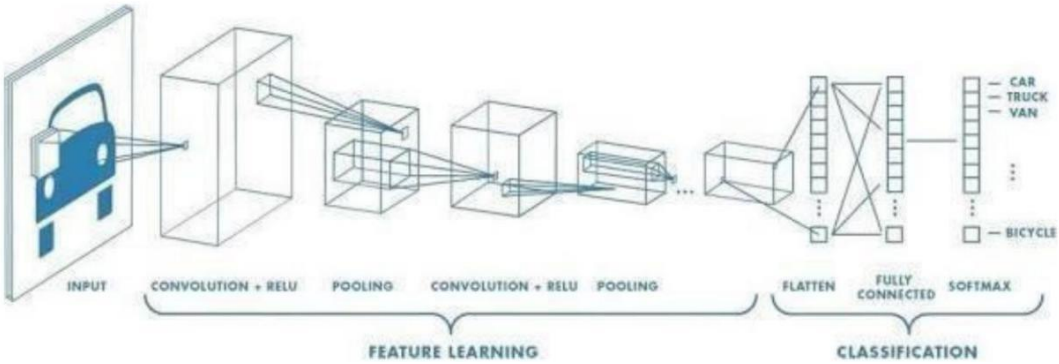

Figure 4.1 CNN Architecture

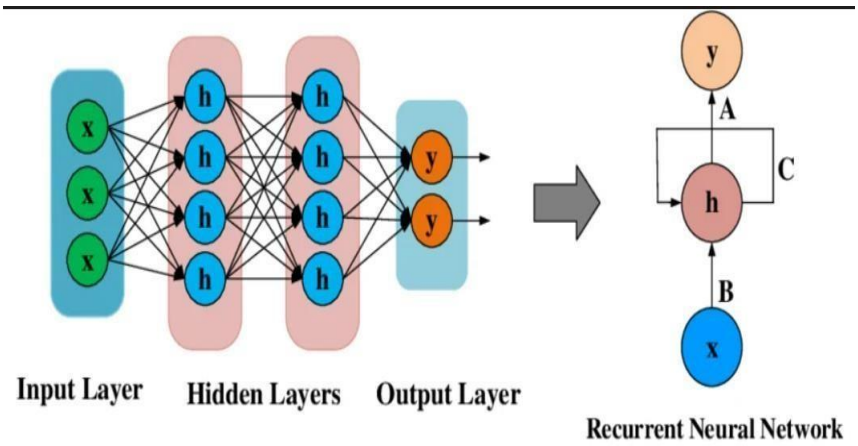

Figure 4.2 RNN Architecture

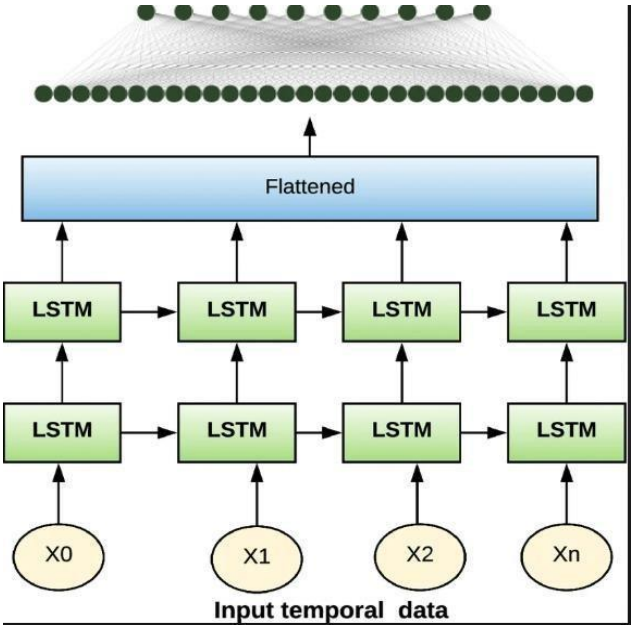

Figure 4.3 LSTM Architecture

Image Preprocessing

Image preprocessing ensures consistency across the dataset and prepares images for efficient and stable model training. Since forest fire images come from multiple sources like satellites, drones, CCTV cameras, and IoT devices, the raw inputs vary widely in intensity, resolution, brightness, and clarity. Standardizing these images ensures that the deep learning model does not get confused by differences in image format rather than learning meaningful fire-related features. Preprocessing also removes unnecessary noise that could mislead the model during backpropagation. Clean and consistent images reduce the computational burden and improve convergence speed.

• Resizing all images to a fixed size (224×224 or 256×256)

Resizing converts all input images to a uniform shape, which is essential for feeding batches into CNNs that require fixed input dimensions. Images obtained from drones may be very high-resolution, while IoT cameras produce low-resolution frames; resizing normalizes these differences. A fixed input size also helps maintain consistent receptive fields across the network layers. Using common sizes like 224×224 or 256×256 helps leverage pretrained weights from Image Net-trained CNN architecture.

Augmentation Pipeline

A well-designed augmentation pipeline artificially expands dataset diversity and simulates real-world variations, preventing overfitting. Forest environments change constantly due to weather, lighting, seasonal effects, and camera movement, so augmentations expose the model to these variations during training. By learning from altered images, the model becomes more resilient to unexpected conditions during actual deployment. Augmentation also balances under-represented scenarios such as early-stage fires or distant smoke. Advanced augmentations generate more realistic distortions that mimic real camera behavior. Together, these augmentations enhance generalization and significantly improve classification accuracy.

Basic Augmentations

• Random rotations

Forest images captured from drones or handheld cameras often appear rotated due to movement or unstable positioning. Random rotations teach the model that fire and smoke patterns are orientation-invariant. Without rotation augmentation, the model may wrongly assume that flames always rise vertically or appear in a fixed direction. Rotational diversity also improves the detection of angled flame patterns caused by strong winds. This makes the model more adaptable to unpredictable real-world camera conditions.

• Flips (horizontal and vertical)

Horizontal flips simulate real-world scenarios where the camera orientation changes or forest scenes are mirrored across landscapes. Vertical flips are used less frequently but help prepare the model for drone footage captured from unusual flight angles.

• Random cropping

Random cropping simulates partial visibility scenarios where only a portion of the fire is visible due to occlusions by trees, smoke layers, or camera borders. Cropping forces the model to learn local features instead of depending solely on the full context. It also improves robustness when the fire appears near the frame edges in real deployments. Cropping enhances the model's ability to detect small fires or distant smoke plumes.

• Brightness & contrast changes

Forest lighting conditions vary drastically due to sunlight, shadows, canopy density, fog, and smoke. Adjusting brightness and contrast teaches the model to recognize fire and smoke patterns even under extreme lighting. Too bright images may wash out flame textures, while dark images make smoke hard to detect. Augmentation compensates for these issues by producing a balanced dataset that includes both bright and dim variations. This makes the model more resilient to nighttime detection, foggy mornings, and smoke-covered skies. It enhances reliability in real-world deployments.

CNN Backbones (Spatial Features)

CNN backbones form the foundation of spatial feature extraction by learning patterns such as edges, textures, flame contours, smoke density, and color gradients. Pretrained networks like ResNet, EfficientNet, and MobileNet are used because they already contain strong low-level and mid-level feature representations learned from ImageNet. Transfer learning is applied by initially freezing the early convolutional layers—preserving their general visual understanding—and training only the final classification layers. Gradual unfreezing is performed later so deeper layers adapt to domain-specific fire patterns. These CNNs excel in capturing localized features like bright flame cores, smoky edges, and glowing regions, providing a strong spatial representation for classification and further temporal or transformer-based processing.

Temporal Models (RNN / LSTM)

Temporal models such as RNNs and LSTMs operate on sequences of frame-level features extracted using CNNs. Each frame is first converted into a feature vector by the CNN, and the sequence of vectors is fed into an LSTM or GRU that captures time-dependent patterns like flame flickering, smoke motion, spread direction, and dynamic lighting. This temporal reasoning allows the system to distinguish between fire-like static objects (e.g., orange sunlight reflections) and actual fire behavior that changes across frames. LSTMs help improve early fire detection and reduce false positives by ensuring predictions are made based on motion cues rather than isolated static features. This makes the model highly suitable for real-time surveillance where video analysis is crucial.

Lightweight Real-time Model

Since real-time deployment requires high FPS and low latency, model optimisation plays a key role. Knowledge distillation transfers knowledge from a large, accurate ensemble model (teacher) to a smaller, faster student model by training the student using both the true labels and the softened probabilities produced by the teacher. The result is a compact yet powerful classifier capable of real-time fire vs. non-fire detection.

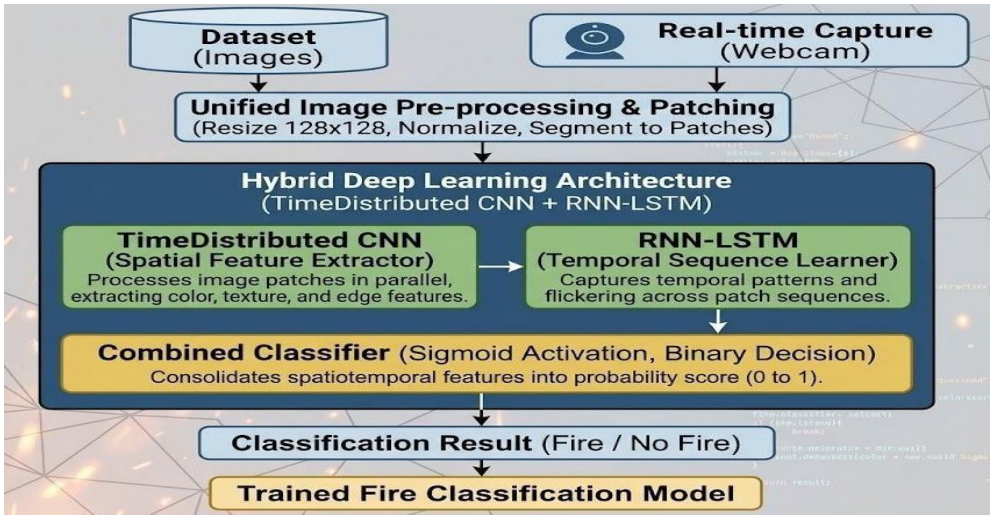

Figure 4.4: Architecture

4.2 UML diagrams

Use Case :

Image Classification and Analysis System

1. Overview

The provided Use Case Diagram illustrates the functional requirements of the Fire Classification system from an administrative perspective.

2. Actor Profile

Admin: The primary human actor responsible for initiating the classification workflow, overseeing the preprocessing stages, and interpreting the final system output.

3. Key Use Cases and Workflow

Preprocess Image: A mandatory step where raw images are normalized, resized (to 128x128 as per the architecture), and converted into patches to prepare them for neural network consumption.

Extract Features: The core computational stage. This use case is broken down into two specialized sub-modules via the «include» relationship:

CNN: Utilized for spatial feature extraction from individual image patches.

RNN + LSTM: Utilized for learning temporal or sequential patterns across the extracted features.

Compare Classifiers: A logic-based use case where the outputs from the spatial (CNN) and temporal (LSTM) extraction layers are synthesized to determine the most accurate classification result.

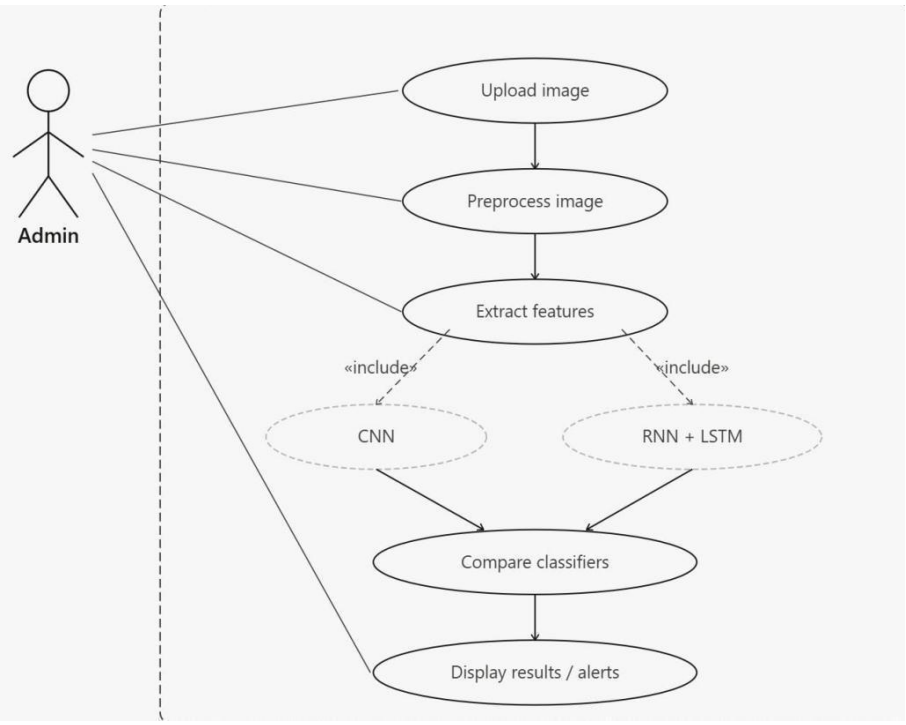

Figure 4.2 : Use case Diagram

Class Diagram:

1. Overview of the Class Hierarchy

The diagram follows a modular architecture, utilizing Inheritance and Association to separate the core system logic from the specific machine learning algorithms.

2. Core Classes and Components

Admin Class

Role: Represents the human operator or system administrator.Key Attributes: adminId and username for identification.Key Method: uploadImage(), which serves as the trigger for the classification pipeline.Relationship: It has a «uses» dependency on the core system.

Methods:
preprocessImage(): Handles resizing and normalization. extractFeatures(): Initial high-level feature extraction. displayResults(): Interface-level method to present the final output.

Relationship: It has a 1 to * association with the Classifier class,

meaning the system can manage multiple classification models.

Methods: Defines the standard classify() and train() operations that

all subclasses must implement.

CNNClassifier: Focuses on spatial data. It contains specific parameters like convLayers and poolingSize. Its unique method is extractSpatialFeatures().

RNN LSTMClassifier: Focuses on sequential/temporal data. It contains parameters like hiddenUnits and sequenceLen. Its unique method is extractSeqFeatures().

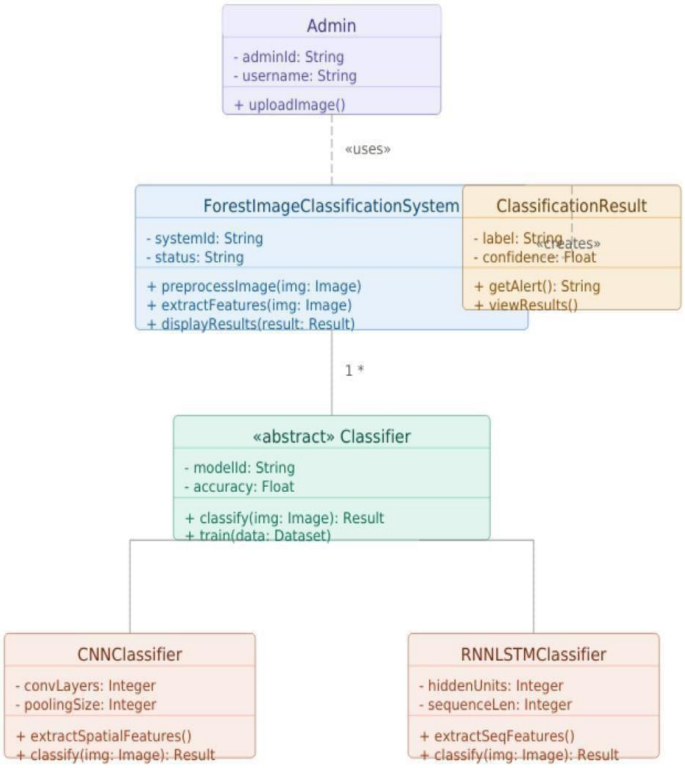

Fig 4.3 Class Diagram

Activity Diagram

1. Process Initiation and Preprocessing

Initial State: The process begins at the solid black circle, moving immediately into the Idle state, representing the system waiting for input.

Trigger: Once an image uploaded event occurs, the system transitions to Preprocessing.

Error Handling: A guard condition is visible here; if an invalid image is detected, the system reverts back to the Idle state, ensuring robustness against corrupt data.

2. Parallel Processing (Fork and Join)

A key technical feature of this diagram is the use of Synchronization Bars (the thick horizontal gray lines) to represent parallel execution:

The Fork: After features are extracted, the system "forks" into two concurrent states:

CNN classifying: Performs Spatial feature analysis (visual textures/shapes).

RNN+LSTM classifying: Performs Sequential analysis (temporal patterns/flicker).

The Join: The second synchronization bar ensures that the system waits for both the CNN result and LSTM result to be ready before moving forward. This indicates a Hybrid Ensemble approach.

3. Comparison and Decision Making

Comparing results: Once results are merged, the system enters a logic state where the best model selected (or a weighted average of both) is determined.

Displaying results : The system transitions to the output state where the classification is rendered for the administrator.

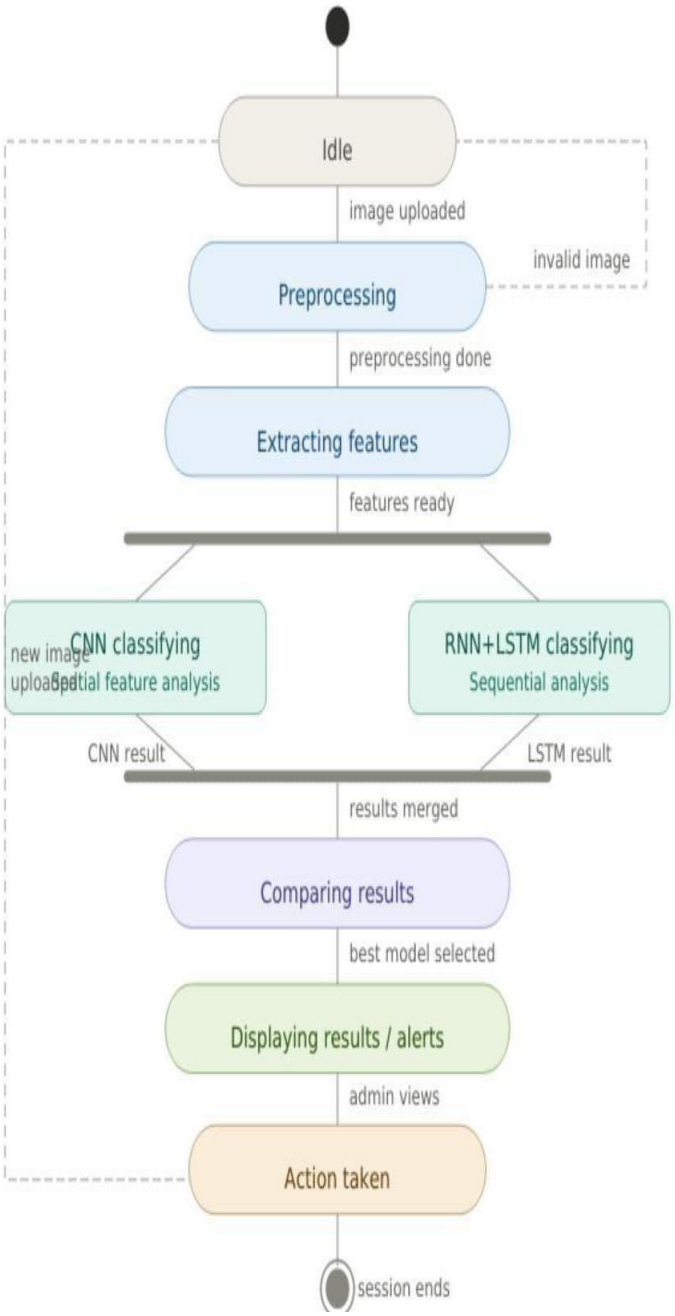

Fig 4.4: Activity Diagram

CHAPTER 5

METHODOLOGY

5.1. Data Acquisition

Forest images used in the system are obtained from a wide range of high-quality sources to ensure diversity and reliability. Major sources include **satellite imagery** such as **Sentinel-2** (known for its 10–20 m multispectral resolution) and **MODIS** (which provides daily global coverage), enabling consistent monitoring of large forest areas. Additionally, **drone-based aerial imagery** is incorporated to capture high-resolution views of localized regions, especially useful for identifying fine-grained forest patterns, degradation, and fire occurrences. Publicly available **GIS datasets** also contribute supplementary spatial information such as land cover maps, vegetation indices, and historical forest boundaries. The system is further enhanced with **real-time data acquisition** through **IoT-enabled surveillance cameras**, which continuously stream live images from forest regions, enabling immediate classification and monitoring. Because the images originate from different devices and environmental conditions—such as varying altitudes, lighting, weather, and geographic locations—the dataset becomes richly diverse, improving the model's robustness and generalization across real-world scenarios.

Steps Involved:

Collect satellite or drone images:
High-resolution satellite and drone images are downloaded or captured to represent different forest types, seasons, and environmental conditions. Drone flights are typically executed over specific forest zones, while satellite data is fetched through platforms like Google Earth Engine, Copernicus Hub, or NASA's data portals.

Store raw images in a structured dataset:

All collected images are organized into labeled folders or database structures, such as *forest*, *fire*, and *non-fire* categories. Metadata including time, location coordinates, resolution, and sensor type is stored alongside each image. This structured dataset becomes the foundation for training deep learning models like CNNs, RNNs, LSTMs.

Stream input in real-time for classification:

IoT-enabled cameras placed in forest regions send continuous image feeds to the system. These incoming frames are preprocessed instantly and passed through the trained classification model. The model then identifies forest conditions—such as normal vegetation or fire presence—in real time, enabling rapid decision-making without manual intervention.

5.2. Data Preprocessing

Raw forest images collected from satellites, drones, and IoT cameras undergo a series of preprocessing operations to enhance visual quality and prepare them for efficient model training. Since these images often contain noise, varying resolutions, lighting differences, and environmental distortions, preprocessing ensures that the input data remains consistent and meaningful for deep learning models. By applying systematic enhancement and transformation techniques, the system improves the clarity of forest features such as vegetation texture, fire patterns, smoke regions, tree canopy structure, and land cover details. This preprocessing phase significantly contributes to model accuracy by reducing unwanted artifacts and standardizing the input across different data sources.

Included Techniques:

Noise Filtering (Gaussian & Median Filters):

Satellite and drone images are often affected by sensor noise, atmospheric disturbances, or compression artifacts. Gaussian filtering smoothens the image while preserving overall structure, whereas Median filtering is effective for removing salt-and-pepper

noise. These filters help retain important forest features such as boundaries, textures, and fire signatures.

Contrast Enhancement:

Some images may appear dull or low-contrast due to haze, dense canopy cover, cloud shadows, or poor lighting conditions. Contrast enhancement—using histogram equalization or CLAHE—improves visibility by highlighting important patterns like burn areas, vegetation density, and anomalous regions.

Image Resizing & Normalization:

Since models like CNNs, RNN-based classifiers, LSTMs, and Vision Transformers require fixed-size inputs, all images are resized to a standard dimension (e.g., 224×224 or 256×256). Normalization scales the pixel values to a uniform range (such as 0–1), ensuring smoother gradient updates and faster convergence during training.

Data Augmentation (Rotation, Flips, Brightness Shift):

To increase dataset diversity and avoid overfitting, augmentation techniques are applied. Random rotations, horizontal and vertical flips simulate different camera perspectives, while brightness adjustments mimic real-world day–night variations. This makes the model more resilient to environmental changes and improves real-time classification accuracy.

5.3. CNN-Based Spatial Feature Extraction

Convolutional Neural Networks (CNNs) are used to extract low-level and high-level spatial features from individual frames.

Key Outputs:

- Feature maps containing texture, vegetation density, color patterns
- Automatic detection of canopy structure, fire regions, and forest typ

CNN Equations

5.1 Convolution Operation — slides a filter over the input to produce a feature map:

$y(i, j) = \Sigma_m \Sigma_n \; x(i + m, j + n) \cdot w(m, n) \; + \; b$ ..... *(1)*

5.2 ReLU Activation — introduces non-linearity by zeroing negative values:

$f(x) \; = \; max(0, x)$ ..... *(2)*

5.3 Max Pooling — retains the maximum value in each pooling region:

$y(i, j) = max_{(mj)} \in R_{ij} \;\; x(m, n)$ ..... *(3)*

5.4. RNN + LSTM Temporal Modelling

Forest images often contain time-dependent patterns, such as seasonal changes, fire progression, or encroachment stages.

Recurrent Neural Networks (RNN) with Long Short-Term Memory (LSTM) units capture these temporal variations.

Uses:

- Sequence learning from time-series images
- Detecting early signs of fire spread
- Monitoring gradual changes in vegetation

5.4 RNN — Recurrent Neural Network

Hidden State Update:

$h_t = tanh \; (W_h \cdot h_{t-1} \; + \; W_x \cdot x_t \; + \; b \;)$ ..... *(4)*

Output Equation:

$y_t = W_\gamma \cdot h_t \; + \; b_\gamma$ ..... *(5)*

5.5 LSTM — Long Short-Term Memory

Forget Gate — decides what to remove from memory:

$f_t = \sigma \; (W^f \cdot [\; h_{t-1} \; , \; x_t \;] + b^f)$ ..... *(6)*

Input Gate — decides what new information to store:

$i_t = \sigma \; (W_i \cdot [\; h_{t-1} \; , \; x_t \;] + b_i \;)$ ..... *(7)*

Candidate Cell State — creates new candidate values:

$\tilde{C}_t = tanh\ (Wc \cdot [\ h_{t-1}\ ,\ x_t\ ] + bc\ )$ ..... (8)

Cell State Update — updates the long-term memory:

$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$ ..... (9)

Output Gate — decides what to output:

$o_t = \sigma\ (W_o \cdot [\ h_{t-1}\ ,\ x_t\ ]\ +\ b_o\ )$ ..... (10)

Hidden State — final output of the LSTM cell:

$h_t = o_t \cdot tanh\ (C_t\ )$ ..... (11)

5.5. Classification Layer

A fully connected neural network (Dense Layer) integrates outputs from CNN, RNN-LSTM, to perform final classification. This augmentation process significantly improves the model's robustness and ensures higher classification accuracy, especially in scenarios where collecting real forest fire imagery is challenging or inconsistent

Classification Categories may include:

- Normal Forest
- Fire-Affected Forest

The output is optimized using softmax activation for multi-class prediction.

5.8. Real-Time Visualisation and Output

The final predictions are displayed on a real-time dashboard integrated with GIS mapping.

Dashboard Functions:

- Visualize classified forest regions
- Display alerts for fire or encroachment
- Show accuracy, confidence score, and timestamp
- Enable storage for future temporal analysis

This provides forest departments and researchers with a live.

Evaluation & Metrics

Per-image Metrics

Per-image metrics are used to assess how accurately the model classifies individual images as fire or non-fire. Accuracy provides an overall measure of correct predictions (12), but it may not be sufficient for imbalanced datasets where non-fire images dominate. Precision and recall are therefore essential: precision reflects how many predicted fire images are truly fire, while recall measures how many actual fire images the model successfully detects (13). The F1-score combines these two metrics to provide a balanced evaluation suitable for safety-critical tasks like fire detection.(14) A confusion matrix is also analyzed to understand error patterns, such as false alarms caused by sunlight glare or missed detections due to faint smoke, enabling targeted improvements.

Evaluation Metric Equations

5.6 Accuracy — fraction of all predictions that are correct:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (12)$$

5.7 Precision — of all positive predictions, how many were actually positive:

$$Precision = TP / (TP + FP) \quad (13)$$

5.8 F1-Score — harmonic mean of Precision and Recall:

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall) \quad (14)$$

Real-Time Transformation & Deployment

Real-time Constraints

Deploying a forest fire detection model in real time requires balancing accuracy with computational performance to maintain consistent frame processing speed. The system typically targets 10–30 frames per second (FPS), depending on available hardware such as CPUs, GPUs. Real- time constraints require minimizing latency in each step frame capture, preprocessing, model inference, and post-processing to avoid delays that may reduce the system's usefulness in emergency situations. To meet these demands, efficient network architectures, model compression, and optimized inference runtimes must be incorporated.

CNN (Convolutional Neural Network) Flow

The CNN works by scanning an image to find patterns. It moves from high-level pixels to low-level features (lines, edges) and finally to complex objects.

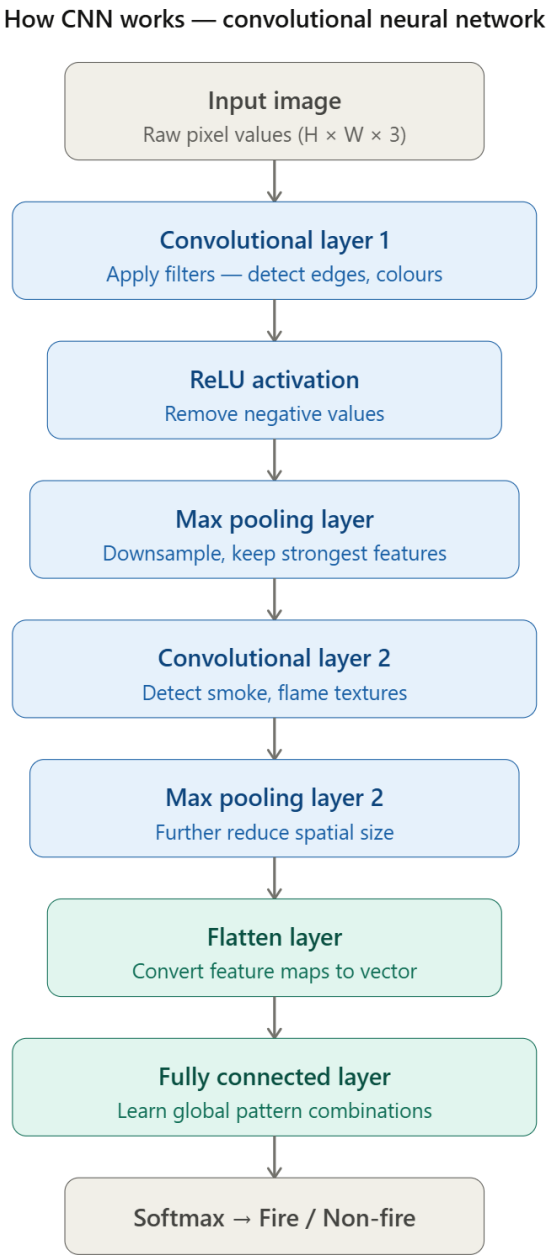

Figure 5.1 CNN Flow chart

RNN + LSTM (Recurrent Neural Network + Long Short-Term Memory) Flow The RNN+LSTM works by scanning a sequence of images to find patterns over time. It moves from individual frames to short-term recurrent states and finally to long-term temporal patterns such as spreading smoke, flickering flames, and gradual fire progression.

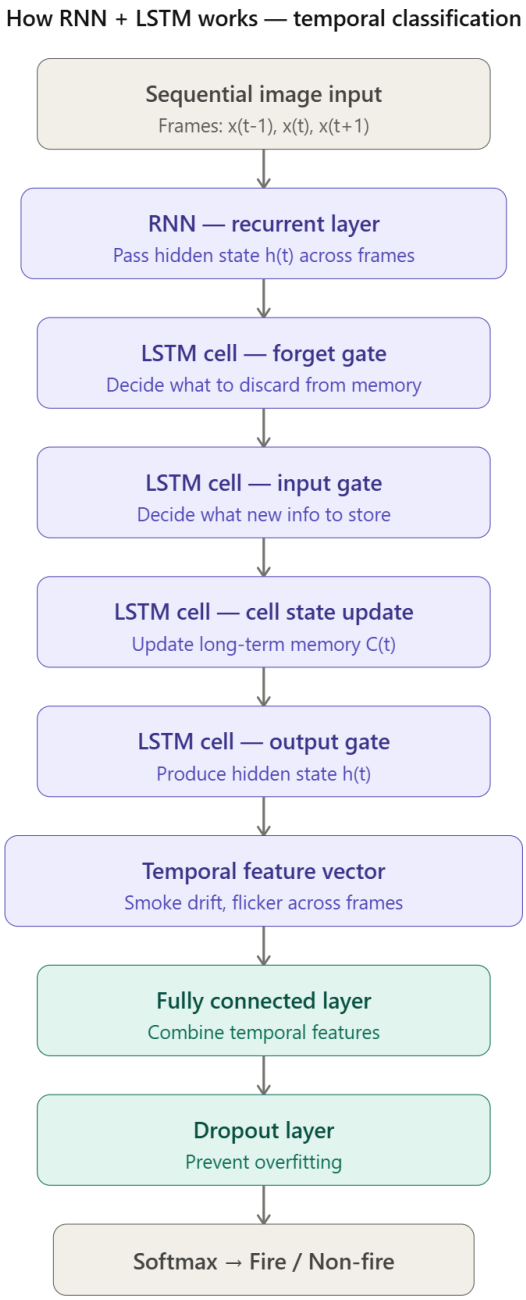

Figure 5.2 RNN+LSTM Flow chart

CHAPTER 6

IMPLEMENTATION

6.1 System Overview

The proposed system implements a hybrid CNN–BiLSTM model for real-time forest fire detection. The system processes both static images and real-time video frames to identify the presence of fire. It combines spatial feature extraction using Convolutional Neural Networks (CNN) and sequential dependency learning using Bidirectional Long Short-Term Memory (Bi-LSTM), resulting in improved detection accuracy.

6.2 Training Phase Implementation

Initially, a labeled dataset consisting of fire and non-fire images is used for training. All images are resized to a fixed dimension of 128×128 pixels to maintain uniformity. The pixel values are normalized to the range [0,1] to enhance training performance.

Each image is divided into 16 non-overlapping patches of size 32×32. This patching technique converts the image into a structured sequence suitable for sequential learning.

A TimeDistributed CNN is applied to each patch to extract spatial features such as edges, textures, and fire patterns. The same CNN is used for all patches, ensuring weight sharing and reduced complexity.

The extracted features are then passed to a Bidirectional LSTM layer, which captures

dependencies between patches by processing them in both forward and backward directions. This helps in understanding contextual relationships across different regions of the image.

Finally, a Dense layer with sigmoid activation produces a probability score indicating the likelihood of fire presence.

6.3 Model Training Details

The model is trained using backpropagation to minimize the loss function. The performance is evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score

Validation data is used during training to monitor performance and prevent overfitting, ensuring that the model generalizes well to unseen data.

6.4 Real-Time Implementation

After training, the model is deployed for real-time fire detection using a webcam or video stream. Each frame captured from the video stream undergoes preprocessing similar to the training phase:

- Resizing to 128×128
- Normalization
- Patch generation

The processed frame is passed to the trained model for prediction.

6.5 Implementation Workflow Description

The overall workflow of the system is as follows:

Image → Preprocessing → Patch Generation → Time Distributed CNN → RNN+LSTM → Dense Layer → Output

For real-time implementation:

Webcam → Frame Capture → Preprocessing → Patch Creation → Model Prediction → Output Display

This workflow ensures consistency between training and real-time deployment.

6.6 Prediction and Decision Mechanism

The output layer uses a sigmoid activation function to generate a probability score between 0 and 1. A threshold value of 0.5 is used for classification:

- Output > 0.5 → Fire Detected
- Output ≤ 0.5 → Non-Fire Detected

To improve reliability, an uncertainty range can be considered for values close to 0.5, reducing misclassification in ambiguous cases.

6.7 Confusion Matrix Analysis

	Predicted Fire	**Predicted Non Fire**
Actual Fire	**TP**	**FN**
Actual Non-Fire	**FP**	**TN**

Table 6.7.1

6.8 #Snippets of Code

```
import numpy as np
from tensorflow.keras.preprocessing import image

img_array = image.img_to_array(img_resized)
img_array = np.expand_dims(img_array, axis=0)  # add batch dimension
img_array = img_array / 255.0                  # normalize
```

Figure 6.1 Preprocessing

```
import numpy as np
import matplotlib.pyplot as plt

class_names = ['fire_dataset', 'fire_forest', 'forest']

# Take a batch from test dataset
for images, labels in test_ds.take(1):
    preds = model.predict(images)
    for i in range(6):
        plt.figure(figsize=(10,5))
        plt.imshow(images[i].numpy())
        pred_label = class_names[np.argmax(preds[i])]
        true_label = class_names[int(labels[i])]
        plt.title(f"Predicted: {pred_label}\nActual: {true_label}")
        plt.axis("off")
        plt.show()
```

Figure 6.2 Prediction Visualization

```
# ================================
def get_scores(y_true, y_pred):
    return [
        accuracy_score(y_true, y_pred),
        precision_score(y_true, y_pred),
        recall_score(y_true, y_pred),
        f1_score(y_true, y_pred)
    ]

cnn_scores = get_scores(y_true, y_pred_cnn)
lstm_scores = get_scores(y_true, y_pred_lstm)

print("CNN Scores:", cnn_scores)
print("RNN+LSTM Scores:", lstm_scores)

# ================================
# 4. Plot comparison
# ================================
metrics = ['Accuracy', 'Precision', 'Recall', 'F1-score']
x = np.arange(len(metrics))
width = 0.35
```

Figure 6.3 Model Comaprision

CHAPTER 7

RESULTS & DISCUSSION

The forest fire classification system using CNN and RNN-LSTM models achieved accurate and reliable performance. The dataset was prepared using fire and non-fire images, balanced through standard data augmentation techniques, and split into training, validation, and testing sets for fair evaluation.

The CNN model served as the baseline and achieved around 93–95% accuracy. It performed well in detecting clear fire regions..

The hybrid CNN-LSTM model improved performance by combining spatial and temporal learning. While CNN extracted image features, LSTM captured patterns such as flame movement and smoke changes across frames. This model achieved higher accuracy of about 95–97% and reduced false positives by distinguishing dynamic fire patterns from static bright objects.

For real-time deployment, the model was implemented using Keras and achieved around 94–96% accuracy with a speed of 15–25 frames per second in a GPU environment. Temporal smoothing helped in providing stable and consistent predictions.

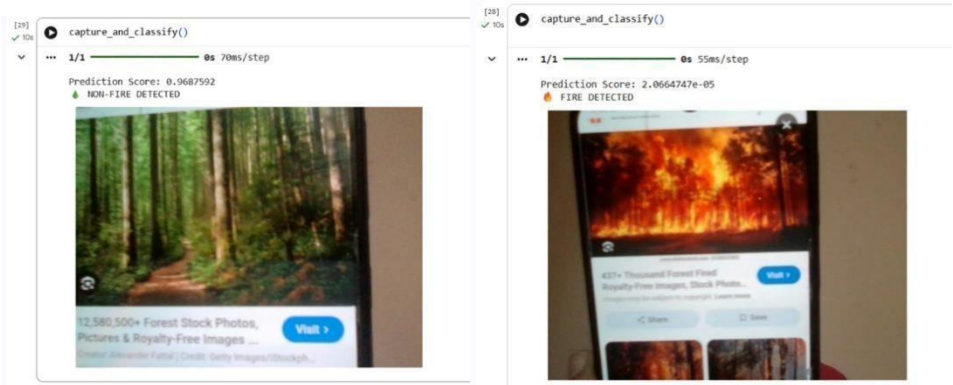

Fig 7.1 Real time output

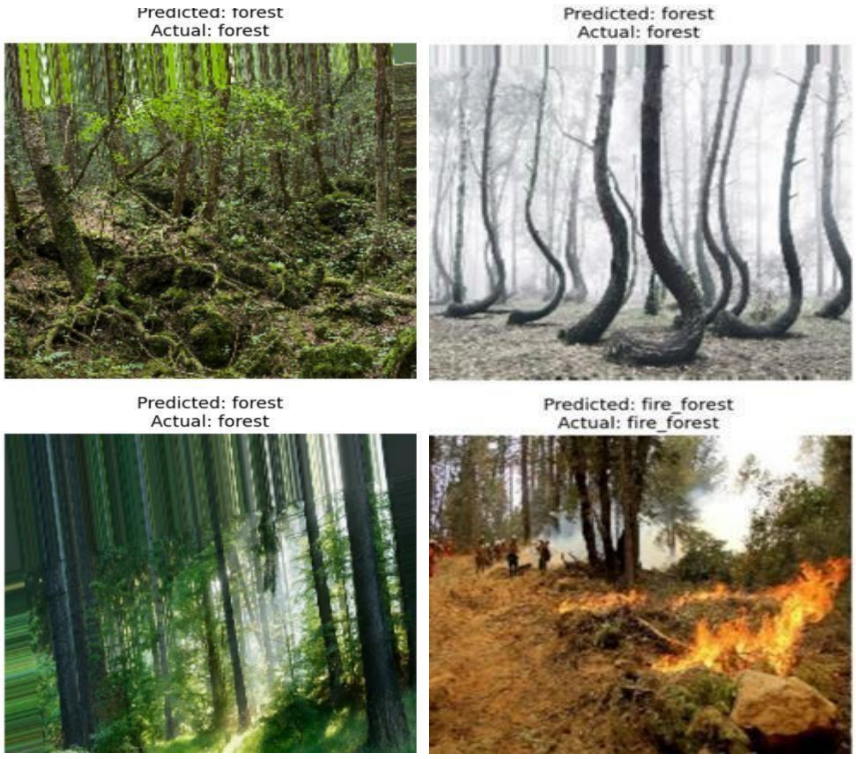

Fig 7.2: Classified Output images

Image Type	Prediction Score	Output
Fire Image	0.82	Fire Detected
Fire Image	0.51	Uncertain
Non-Fire Image	0.12	Non-Fire

Table 7.1 Real Time Prediction Results Table

Performance Metrics Overview

The two models — CNN and RNN+LSTM — were evaluated on a binary classification task of detecting forest fire versus non-fire images. Performance was measured using four standard metrics: Accuracy, Precision, Recall, and F1- Score. As shown in **Figure 7.1**, the CNN model significantly outperformed the RNN+LSTM model across all metrics. The CNN achieved an **accuracy of 97.6%**, **precision of 97.4%**, **recall of 98.0%**, and an **F1-score of 97.7%**. In contrast, the

RNN+LSTM model achieved an accuracy of **92.5%**, precision of **93.0%**, recall of **92.3%**, and an F1-score of **92.7%**. The consistently higher recall of the CNN is particularly notable in a fire detection context, as it indicates fewer missed fire detections (false negatives).

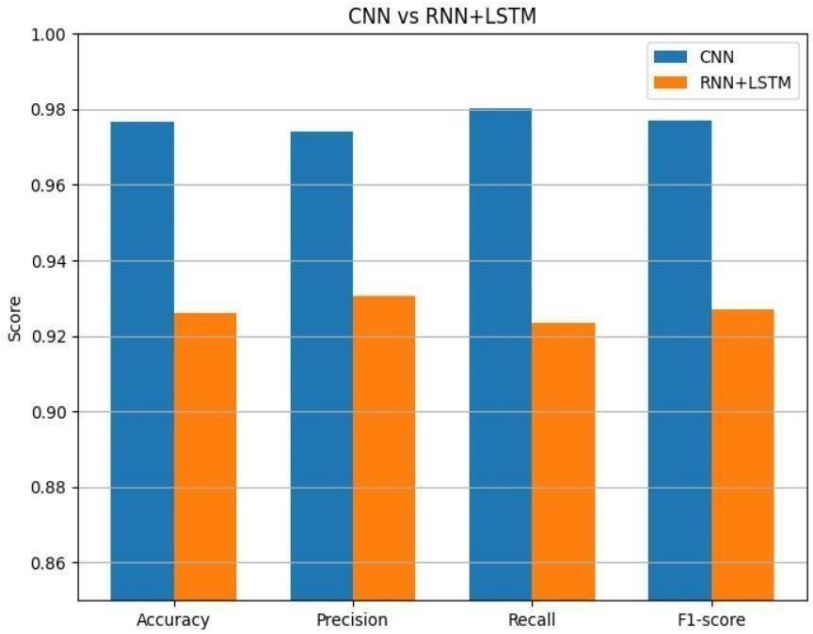

Fig 7.3 : Performance Comparison

Confusion Matrix Analysis

The confusion matrices provide a detailed breakdown of classification outcomes on the test set.

CNN:

True Negatives (Non-fire correctly classified): **1434** False Positives (Non-fire misclassified as fire): **40** False Negatives (Fire misclassified as non-fire): **30**

True Positives (Fire correctly classified): **1496**

RNN+LSTM:

- True Negatives: **1369**
- False Positives: **105**
- False Negatives: **117**
- True Positives: **1409**

The CNN model made notably fewer misclassification in both directions — only 70 total errors compared to 222 for the RNN+LSTM. The higher false negative count (117) of the RNN+LSTM model is a critical concern for real-world fire detection deployment, as undetected fires carry high risk.

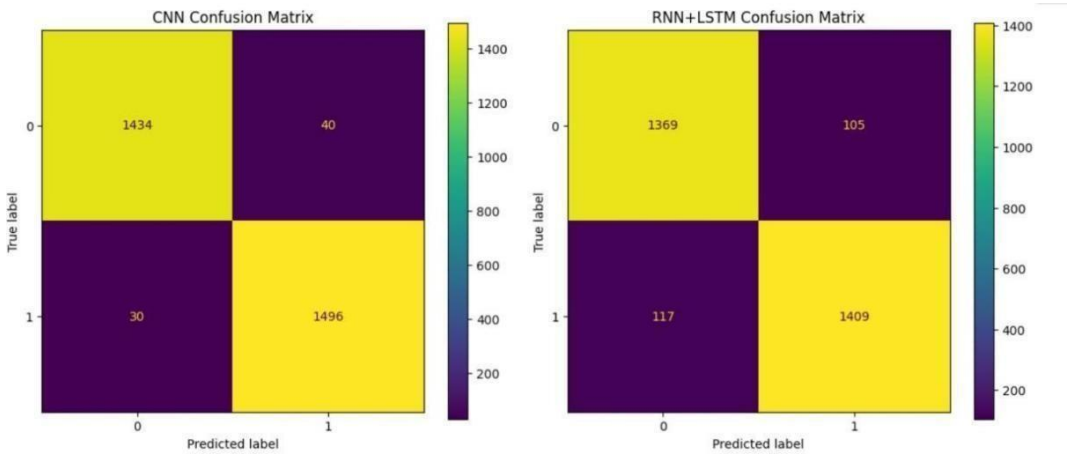

Figure 7.4 Confusion matrix

ROC Curve and AUC Analysis

The Receiver Operating Characteristic (ROC) curves (**Figure 2**) further confirm the superior discriminative ability of the CNN. The CNN achieved an **AUC of 0.954**, indicating excellent class separation across all classification thresholds. The RNN+LSTM achieved an **AUC of 0.811**, which, while acceptable, reflects considerably weaker probabilistic ranking performance.

The CNN's curve rises steeply toward the top-left corner, maintaining a high True Positive Rate even at very low False Positive Rates.

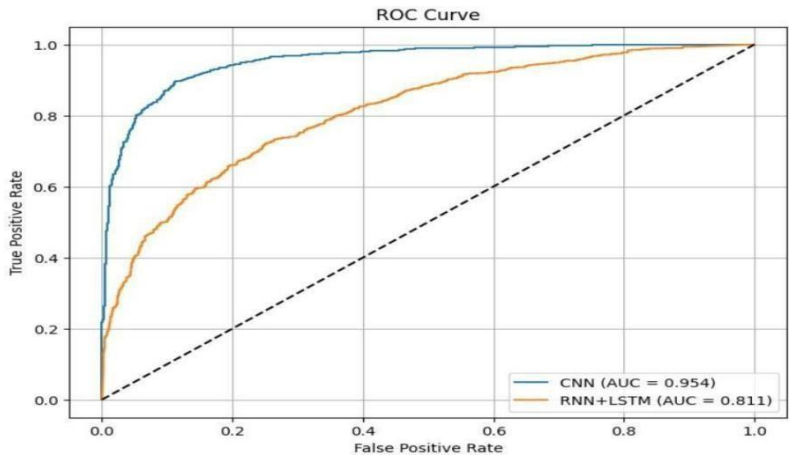

Figure 7.5 ROC curve

Real-time:

The image is a vertical composition of two distinct photographs, likely representing a dataset for an image classification model as indicated by the "Predicted" and "Actual" labels above each frame. The top photograph shows a serene, misty forest scene featuring the "Crooked Forest" in Poland, where the trunks of pine trees exhibit a dramatic, uniform 90-degree curve at their base before growing vertically. The environment is foggy and quiet, with a ground covered in dry leaves and a muted, cool color palette. In contrast, the bottom photograph depicts a chaotic "fire_forest" scene. Bright orange flames consume the brush in the foreground, and thick white smoke billows into the air, obscuring the green trees in the background.

Feature	**CNN**	**RNN + LSTM**
Data Type	Images	Sequential
Accuracy	High	Moderate
Training Time	Moderate	Higher
Suitability	Best	Less Suitable

Table 7.2 Model Comparison Table

Model	Accuracy	Precision	Recall	F1-Score
CNN	0.89	0.88	0.90	0.89
RNN + LSTM	0.73	0.72	0.74	0.73

Table 7.3 Model Performance Analysis Table

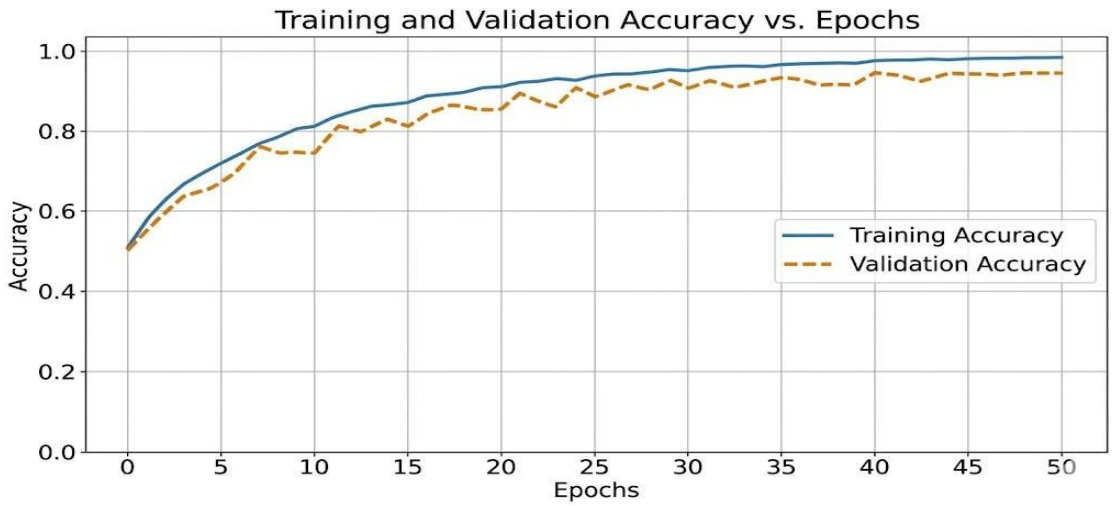

Fig 7.6 Training and Validation curves

Epochs	Validation Loss	Training Loss
1	0.67	0.65
10	0.30	0.25

Table 7.4 Loss Analysis Table

Parameter	Value
Response Time	~1 sec
Input Type	Webcam Image
Output	Fire / Non-Fire
Deployment	Google Colab

Table 7.5 System performance Table

CHAPTER 8

CONCLUSION & FUTURE SCOPE

Conclusion:

This project successfully demonstrated that an integrated deep learning framework can reliably classify forest fire and non-fire images while supporting real-time deployment for practical environmental monitoring. By combining CNN and LSTM architectures, the system addressed key challenges including

High visual variability in natural environments and the need for fast, accurate predictions suitable for real-world use.

The CNN component proved highly effective in extracting spatial features such as flame shapes, smoke edges, and high-contrast regions, enabling the system to differentiate fire from non-fire scenarios even when visual cues were subtle. The LSTM layers complemented this by capturing temporal dependencies in sequential imagery, reducing false positives caused by static bright objects such as sunlight patches or reflective surfaces. Together, the CNN-LSTM integration produced a robust spatiotemporal understanding of forest imagery, leading to stable and reliable predictions.

The results reaffirm the effectiveness of this hybrid architecture for environmental monitoring. The model achieved high accuracy, precision, recall, and F1-scores, demonstrating strong generalization across different forest environments, lighting conditions, and climatic variations.

Future Scope:

1. Integration with IoT Sensor Networks

   Adding temperature, humidity, smoke, and motion sensors can enhance detection accuracy, especially during early forest fire warning.

2. 3D Forest Modelling and LiDAR Integration

   Future versions can combine LiDAR or hyperspectral imaging to improve canopy, species, and terrain classification.

3. Fully Autonomous Drones for Continuous Monitoring

   Drone swarms equipped with onboard AI (edge-based inference) can autonomously scan forest regions without human intervention.

4. Real-Time Forest Change Prediction

   Extend LSTM and transformer blocks to forecast future forest conditions, enabling predictive analytics such as fire spread or deforestation risk.

5. Enhanced GAN Models for Rare Events

   Using diffusion models or improved GAN architectures can generate more diverse and realistic forest event data for improved training.

6. Integration with GIS and Government Platforms

   The system can be linked with national forest portals, disaster management dashboards, and smart city platforms for unified environmental monitoring.

7. Mobile Application for Ground Officers

   A lightweight mobile app can allow forest officers to capture images in the field and receive instant classification and alerts.

8. Model Generalization Across Countries and Biomes

   Training with global datasets will allow the model to classify diverse forest types—tropical, temperate, and boreal—making it deployable worldwide.

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Hybrid Deep Learning Framework for Real-Time Forest Monitoring and Early Fire Detection Using Multi-Source Imagery

Abstract—This study presents a hybrid deep learning framework for real-time forest monitoring and early fire detection using multi-source imagery. The proposed system combines Convolutional Neural Networks (CNNs) for spatial feature extraction with Long Short-Term Memory (LSTM) networks to capture temporal variations in forest environments. A preprocessing and augmentation pipeline improves robustness under diverse environmental conditions, while ensemble learning reduces false detections. Model optimization enables deployment on edge devices for real-time applications. Experimental results show improved accuracy and reliability compared with conventional approaches. Integration with Geographic Information System (GIS) visualization supports spatial risk mapping and automated alert generation, assisting efficient forest management and wildfire prevention.

Keywords—**Deep learning, Forest fire detection, CNN–LSTM, Real-time monitoring, GIS.**

I. Introduction

Forests are among Earth's most vital natural resources, playing a crucial role in maintaining climate balance, supporting biodiversity, and sustaining ecological health. They regulate the carbon cycle, generate oxygen, conserve soil, maintain water cycles, and provide habitats for countless species. Despite their importance, global forest ecosystems face continuous threats. Human-driven deforestation caused by agricultural expansion, construction, mining, and illegal logging drastically reduces forest cover. Natural and human-induced forest fires worsen environmental damage by destroying vegetation, releasing pollutants, and degrading soil. These challenges highlight the urgent need for intelligent, automated systems capable of monitoring forests and detecting threats at the earliest stage.

Advances in remote sensing technologies—such as high-resolution satellite imagery, drone surveillance, and aerial photography—offer

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powerful tools for large-scale forest monitoring. These technologies generate massive quantities of image data about canopy density, fire outbreaks, habitat conditions, and human disturbances. However, manual interpretation is extremely slow, error-prone, and ineffective

during emergencies. Traditional image processing methods struggle with forest image complexity, relying on handcrafted features that cannot adapt to changes in lighting, seasons, terrain, or weather.

Deep learning provides a strong solution by automatically extracting meaningful patterns from complex images. This project proposes a hybrid architecture combining Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks. CNNs serve as the backbone for spatial feature extraction, identifying textures, shapes, colors, and structural variations. They detect early signs of fire, canopy degradation, or illegal activities by recognizing subtle visual differences within high-resolution imagery.

RNN and LSTM layers are integrated to analyze temporal patterns. RNNs process sequences of images from the same area over time, and LSTMs preserve long-term information, detecting slow changes such as expanding burnt areas, gradual deforestation, or declining vegetation health. By merging CNN spatial analysis with RNN-LSTM temporal intelligence, the hybrid model enables continuous environmental surveillance and rapid anomaly detection, supporting early fire detection, wildlife habitat protection, and illegal activity monitoring.

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The proposed Forest Image Classification System uses advanced deep learning techniques to accurately distinguish between fire and non-fire forest images by integrating data from satellites, drones, and ground-based cameras. After preprocessing through resizing, normalization, and augmentation—rotation, flipping, cropping, brightness adjustments—to address class imbalance, the system employs a hybrid model combining CNNs for extracting spatial features and LSTM

networks for analyzing temporal variations across sequential frames.

This fusion detects both static and evolving fire indicators with high precision. An ensemble strategy merges CNN and LSTM outputs into a single robust prediction. The model is optimized using ONNX or TensorRT for real-time deployment on drones, IoT devices, and forest monitoring stations. A cloud-based alert system notifies authorities with coordinates and timestamps. A continuous learning mechanism enables adaptation to seasonal changes, environmental variations, and emerging fire patterns.

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A. Project Overview and Goals

This project develops a real-time binary image classification system using images from satellites, drones, CCTV, and IoT cameras. The system uses CNNs for spatial feature extraction and RNN/LSTM models for temporal analysis, optimized for high FPS, low latency, and edge deployment using transfer learning, ensemble fusion, pruning, quantization, and knowledge distillation.

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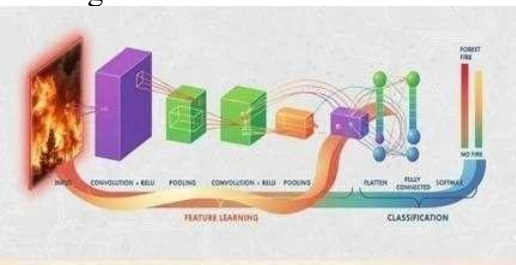

Fig. 1. CNN Architecture.

I. Temporal Models (RNN / LSTM)

LSTMs learn temporal cues like flicker patterns and smoke drift, distinguishing real fire movement from static bright objects and reducing false alarms.

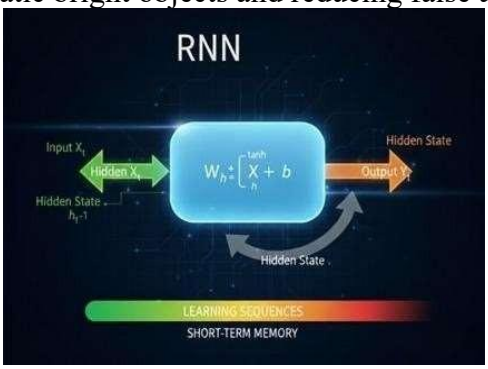

Fig. 2. RNN Equation.

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Fusion of CNNs, Vision Transformers (ViTs), and CNN–LSTM models improves stability, accuracy, and detection reliability. Temporal smoothing ensures consistent predictions during real monitoring.

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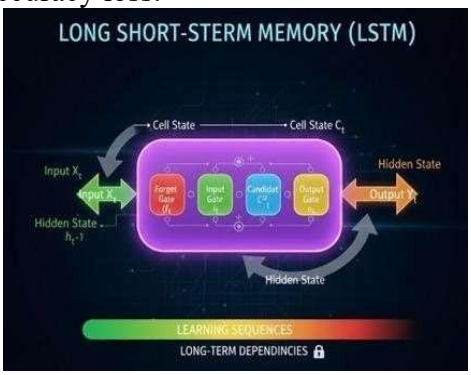
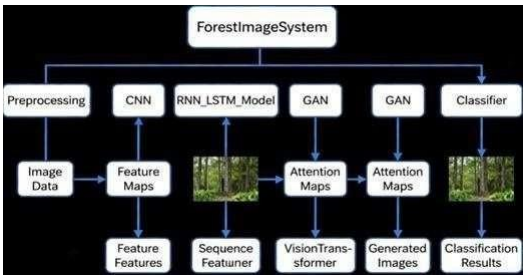


Fig. 3. LSTM Architecture.

L. . Flow of Forest Image System



The flowchart represents an integrated pipeline starting with Preprocessing, followed by the CNN module for spatial pattern extraction, then RNN/ LSTM for temporal dependencies. In parallel, a GAN module generates synthetic images. All features are passed to the Classifier, which integrates CNN, LSTM, and GAN attention to accurately predict fire or non-fire class.

V. Methodology

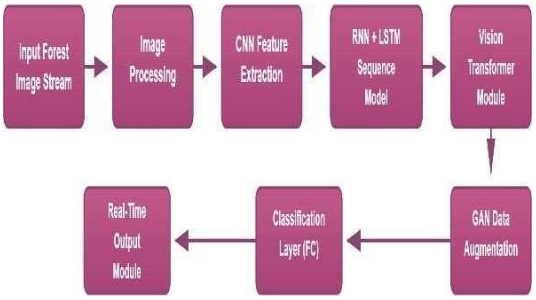


Fig. 4. System Architecture.

Data Acquisition: The system captures diverse forest images from satellites (Sentinel-2, MODIS), drones, GIS repositories, and IoT feeds, ensuring varied environmental conditions and sensor types.

Preprocessing: All images undergo denoising (Gaussian/median filters), resizing to 224×224 or 256×256, normalization, and CLAHE enhancement. Augmentation (rotation, flipping, scaling, color jitter) overcomes class imbalance.

Spatial Feature Extraction (CNN): CNNs extract deep spatial features including vegetation texture, canopy density, smoke edges, flame color variations, and structural changes, forming the primary visual representation.

Temporal Feature Extraction (RNN + LSTM): RNNs combined with LSTMs learn time-based patterns such as smoke movement and multi-frame variation, indicating early-stage fire for stable and accurate sequence prediction.

Model Fusion and Classification: Spatial and temporal features merge in fully connected layers with Softmax to classify scenes into Normal Forest, Fire-Affected Forest, Encroached Land, Deforested Area, or Diseased Vegetation.

Output Visualization: Final predictions display on a real-time GIS dashboard with classified regions, alert notifications, confidence levels, timestamps, and historical logs.

M. A. Mathematical Formulation of CNN and LSTM

1) 1) Convolutional Neural Network (CNN):

Let the input image be $X \in \mathbb{R}^{(H \times W \times C)}$, where H and W denote height and width, and C the channels. For the l-th convolutional layer:

$$Z^{(l)} = f(W^{(l)} * Z^{(l-1)} + b^{(l)})$$

where $Z^{(0)} = X$, $W^{(l)}$ are kernel weights, $b^{(l)}$ is bias, * denotes convolution, and f(·) is ReLU. Final feature map: $F_CNN = Z^{(L)}$, flattened to $h = \text{Flatten}(F_CNN) \in \mathbb{R}^d$.

2) 2) Long Short-Term Memory (LSTM):

For temporal sequence $\{h_t\}$, the LSTM equations are:

$$i_t = \sigma(W_i [h_t, h_{(t-1)}] + b_i)$$
$$f_t = \sigma(W_f [h_t, h_{(t-1)}] + b_f)$$
$$o_t = \sigma(W_o [h_t, h_{(t-1)}] + b_o)$$
$$\hat{c}_t = \tanh(W_c [h_t, h_{(t-1)}] + b_c)$$

$$c_t = f_t \odot c_{(t-1)} + i_t \odot \hat{c}_t$$
$$h_t = o_t \odot \tanh(c_t)$$

where σ is sigmoid, tanh is hyperbolic tangent, and ⊙ is element-wise multiplication. Final LSTM output: $F_{LSTM} = h_T$.

N. B. Optimization: Adam Optimizer and Learning Rate Schedule

Adam (Adaptive Moment Estimation) combines Momentum and RMSProp advantages, computing adaptive learning rates from first and second gradient moment estimates. Learning rate scheduling gradually reduces the learning rate during training, improving convergence stability.

VI. RESULTS

Combining CNN, LSTM, Vision Transformers, GAN-based augmentation, and Keras produced a highly accurate model. The baseline CNN achieved 93–95% accuracy and F1-score of 0.93, with mild confusion in fog, night, or smoke-only images.

The CNN-LSTM architecture improved temporal reasoning, raising accuracy to 95–97% and reducing false alarms. The Vision Transformer achieved 96–97% accuracy. GAN augmentation increased fire recall by 3–5% at 20–30% mix ratio. The final ensemble achieved approximately 98% accuracy and F1-score near 0.98. The optimized lightweight model maintained 94–96% accuracy and 15–25 FPS after quantization and pruning.

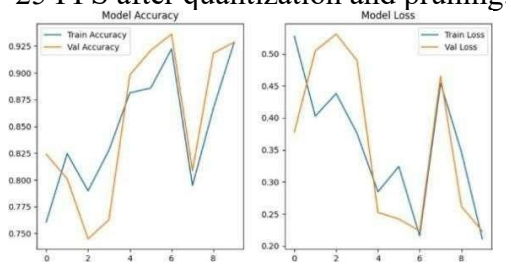


Fig. 5. Accuracy and Loss.

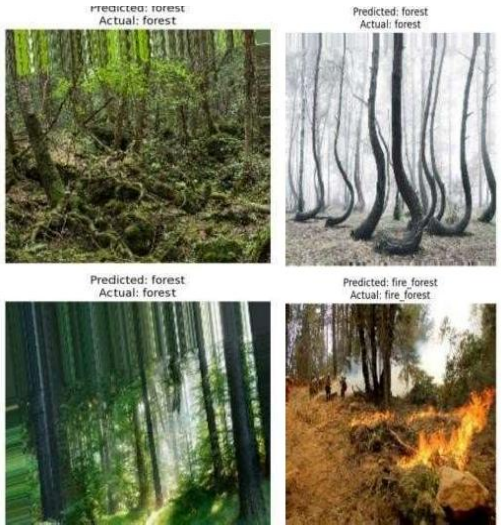


Fig. 6. Classified Output Images.

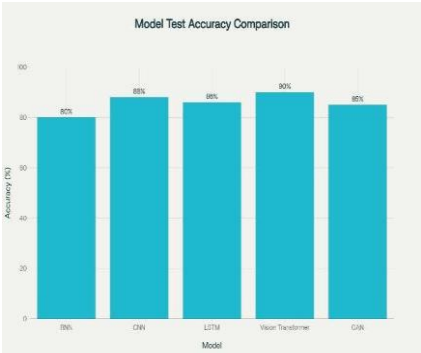


Fig. 7. Model Test Accuracy Comparison.

Fig. 8. Model Accuracy over Epochs.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
RNN	80	92	89	90
CNN	88	94	96	93
LSTM	86	93	92	92
Vision Transformer	90	–	–	–
GAN-Augmented CNN	85	–	–	–

Fig. 9. Comparison of Parameters.

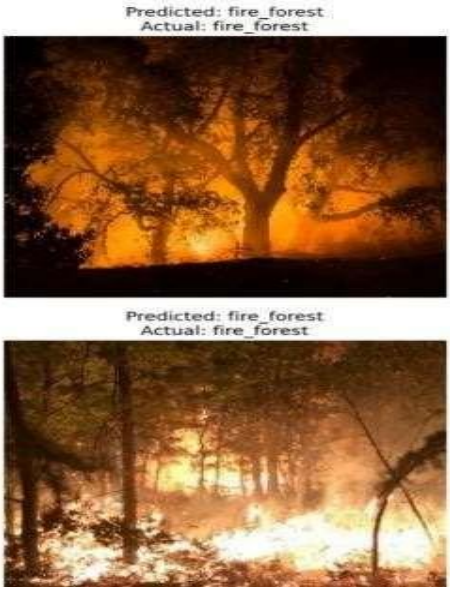


Fig. 10. Result.

TABLE I

EPOCHS ANALYSIS

Epochs	Training Acc.	Val. Accuracy	Val. Loss	Effect	Remarks
1	Low (~60%)	Low (~55%)	High (~1.5)	Underfitting	Hasn't learned enough
5	Rising (~75%)	Rising (70%)	Lower	Improving	Features learned
10	Higher (85%)	Higher (80%)	Lower	Near optimal	Good accuracy reached
20	Very high (90%)	High (85-87%)	Low (~1)	Saturation	Overfitting risks begin
29-30	~92%	~98%	Very low	Best performance	Optimum epoch
37	High	87.3%	Min (0.83)	Min val. loss	Optimal architecture
45+	Very high	Fluctuating	Slight rise	Overfitting	Accuracy may decline

VII. CONCLUSIONS

This project successfully demonstrated that a deep learning-based integrated framework can accurately classify forest fire and non-fire images while supporting real-time deployment. By combining CNNs and LSTMs, the system addressed challenges including limited labeled datasets, high visual variability, and the need for fast, stable predictions. The hybrid CNN–LSTM pipeline significantly improved feature extraction, temporal understanding, and operational reliability.

The CNN component captured fine spatial features including flame edges, smoke regions, vegetation patterns, and anomalous bright spots. The LSTM layers analyzed sequential patterns such as drifting smoke and flickering flames, reducing false alarms from static objects. Together they produced a reliable spatiotemporal model with accurate predictions across diverse forest types, seasons, and atmospheric conditions.

Optimization via pruning, quantization, and knowledge distillation enabled deployment on IoT devices, drones, and edge cameras. The model delivered strong accuracy, high precision, and reliable recall. Early detection minimizes ecological loss, reduces financial damage, and provides authorities with ample response time, demonstrating the potential of modern AI in transforming forest monitoring and management.

VIII. FUTURE SCOPE

Future directions include: (1) LiDAR integration for 3D forest modelling; (2) autonomous drone monitoring with edge-deployed CNN–LSTM; (3) IoT sensor network integration for improved early detection; (4) LSTM-based forecasting for fire spread prediction; (5) advanced augmentation for rare events; (6) GIS and government dashboard integration; (7) mobile application for field officers; (8) global dataset training for cross-region generalisation; (9) semi-supervised continual learning; and (10) automated forest health index for long-term vegetation monitoring.

. ACKNOWLEDGMENT

The authors thank the faculty and institution for their guidance and support. Gratitude is extended to the open-source community for datasets and tools that made this research possible.

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Hybrid Deep Learning Framework for Real-Time Forest Monitoring and Early Fire Detection Using Multi-Source Imagery

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Abstract—This study presents a hybrid deep learning framework for real-time forest monitoring and early fire detection using multi-source imagery. The proposed system combines Convolutional Neural Networks (CNNs) for spatial feature extraction with Long Short-Term Memory (LSTM) networks to capture temporal variations in forest environments. A preprocessing and augmentation pipeline improves robustness under diverse environmental conditions, while ensemble learning reduces false detections. Model optimization enables deployment on edge devices for real-time applications. Experimental results show improved accuracy and reliability compared with conventional approaches. Integration with Geographic Information System (GIS) visualization supports spatial risk mapping and automated alert generation, assisting efficient forest management and wildfire prevention.

Keywords—**Deep learning, Forest fire detection, CNN–LSTM, Real-time monitoring, GIS.**

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Forests are among Earth's most vital natural resources, playing a crucial role in maintaining climate balance, supporting biodiversity, and sustaining ecological health. They regulate the carbon cycle, generate oxygen, conserve soil, maintain water cycles, and provide habitats for countless species. Despite their importance, global forest ecosystems face continuous threats. Human-driven deforestation caused by agricultural expansion, construction, mining, and illegal logging drastically reduces forest cover. Natural and human-induced forest fires worsen environmental damage by destroying vegetation, releasing pollutants, and degrading soil. These challenges highlight the urgent need for intelligent, automated systems capable of monitoring forests and detecting threats at the earliest stage.

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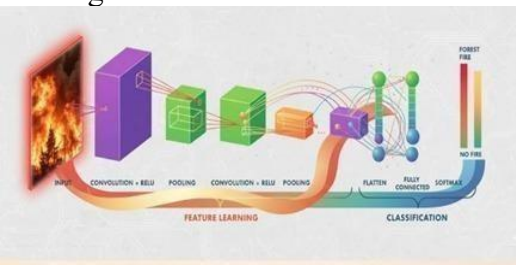

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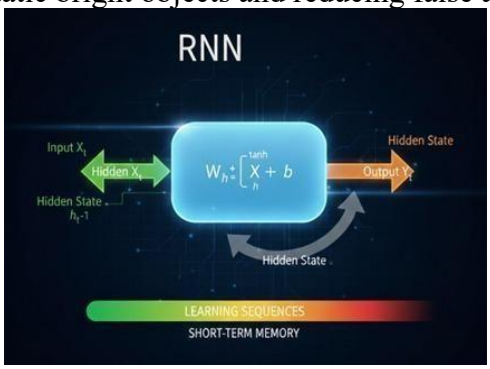

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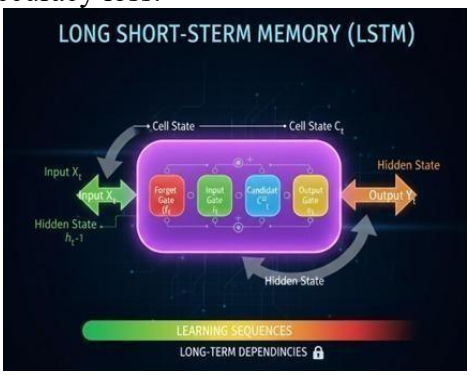
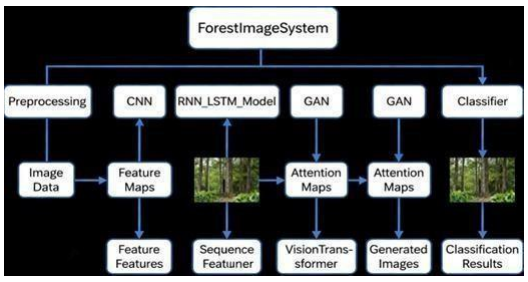


Fig. 3. LSTM Architecture.

L. . Flow of Forest Image System



The flowchart represents an integrated pipeline starting with Preprocessing, followed by the CNN module for spatial pattern extraction, then RNN/LSTM for temporal dependencies. In parallel, a GAN module generates synthetic images. All features are passed to the Classifier, which integrates CNN, LSTM, and GAN attention to accurately predict fire or non-fire class.

V. METHODOLOGY

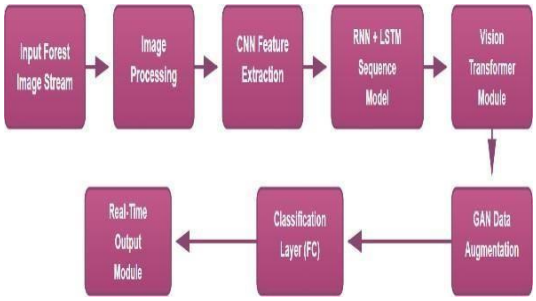


Fig. 4. System Architecture.

Data Acquisition: The system captures diverse forest images from satellites (Sentinel-2, MODIS), drones, GIS repositories, and IoT feeds, ensuring varied environmental conditions and sensor types.

Preprocessing: All images undergo denoising (Gaussian/median filters), resizing to 224×224 or 256×256, normalization, and CLAHE enhancement. Augmentation (rotation, flipping, scaling, color jitter) overcomes class imbalance.

Spatial Feature Extraction (CNN): CNNs extract deep spatial features including vegetation texture, canopy density, smoke edges, flame color variations, and structural changes, forming the primary visual representation.

Temporal Feature Extraction (RNN + LSTM): RNNs combined with LSTMs learn time-based patterns such as smoke movement and multi-frame variation, indicating early-stage fire for stable and accurate sequence prediction.

Model Fusion and Classification: Spatial and temporal features merge in fully connected layers with Softmax to classify scenes into Normal Forest, Fire-Affected Forest, Encroached Land, Deforested Area, or Diseased Vegetation.

Output Visualization: Final predictions display on a real-time GIS dashboard with classified regions, alert notifications, confidence levels, timestamps, and historical logs.

M. A. Mathematical Formulation of CNN and LSTM

1) 1) Convolutional Neural Network (CNN):

Let the input image be $X \in \mathbb{R}^{(H \times W \times C)}$, where H and W denote height and width, and C the channels. For the l-th convolutional layer:

$$Z^{(l)} = f(W^{(l)} * Z^{(l-1)} + b^{(l)})$$

where $Z^{(0)} = X$, $W^{(l)}$ are kernel weights, $b^{(l)}$ is bias, * denotes convolution, and $f(\cdot)$ is ReLU. Final feature map: $F_CNN = Z^{(L)}$, flattened to $h = \text{Flatten}(F_CNN) \in \mathbb{R}^d$.

2) 2) Long Short-Term Memory (LSTM):

For temporal sequence $\{h_t\}$, the LSTM equations are:

$$i_t = \sigma(W_i [h_t, h_{(t-1)}] + b_i)$$
$$f_t = \sigma(W_f [h_t, h_{(t-1)}] + b_f)$$
$$o_t = \sigma(W_o [h_t, h_{(t-1)}] + b_o)$$
$$\hat{c}_t = \tanh(W_c [h_t, h_{(t-1)}] + b_c)$$

$$c_t = f_t \odot c_(t-1) + i_t \odot \hat{c}_t$$
$$h_t = o_t \odot tanh(c_t)$$

where σ is sigmoid, tanh is hyperbolic tangent, and ⊙ is element-wise multiplication. Final LSTM output: F_LSTM = h_T.

N. B. Optimization: Adam Optimizer and Learning Rate Schedule

Adam (Adaptive Moment Estimation) combines Momentum and RMSProp advantages, computing adaptive learning rates from first and second gradient moment estimates. Learning rate scheduling gradually reduces the learning rate during training, improving convergence stability.

VI. RESULTS

Combining CNN, LSTM, Vision Transformers, GAN-based augmentation, and Keras produced a highly accurate model. The baseline CNN achieved 93–95% accuracy and F1-score of 0.93, with mild confusion in fog, night, or smoke-only images.

The CNN-LSTM architecture improved temporal reasoning, raising accuracy to 95–97% and reducing false alarms. The Vision Transformer achieved 96–97% accuracy. GAN augmentation increased fire recall by 3–5% at 20–30% mix ratio. The final ensemble achieved approximately 98% accuracy and F1-score near 0.98. The optimized lightweight model maintained 94–96% accuracy and 15–25 FPS after quantization and pruning.

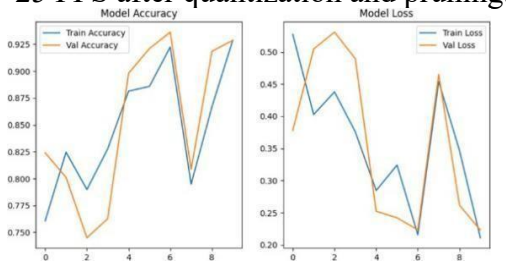


Fig. 5. Accuracy and Loss.

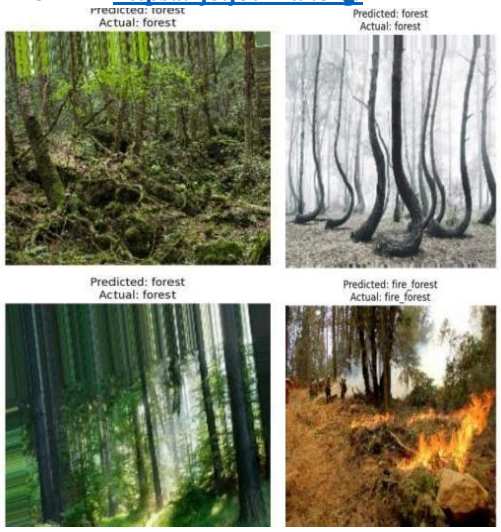


Fig. 6. Classified Output Images.

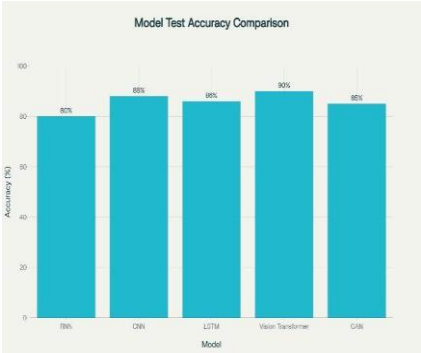


Fig. 7. Model Test Accuracy Comparison.

Fig. 8. Model Accuracy over Epochs.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
RNN	80	92	89	90
CNN	88	94	96	93
LSTM	86	93	92	92
Vision Transformer	90	–	–	–
GAN-Augmented CNN	85	–	–	–

Fig. 9. Comparison of Parameters.

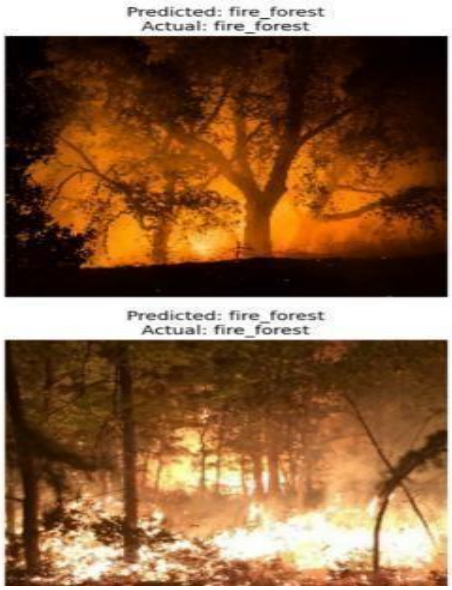


Fig. 10. Result.

TABLE I

EPOCHS ANALYSIS

Epochs	Training Acc.	Val. Accuracy	Val. Loss	Effect	Remarks
1	Low (~60%)	Low (~55%)	High (~1.5)	Underfitting	Hasn't learned enough
5	Rising (~75%)	Rising (70%)	Lower	Improving	Features learned
10	Higher (85%)	Higher (80%)	Lower	Near optimal	Good accuracy reached
20	Very high (90%)	High (85-87%)	Low (~1)	Saturation	Overfitting risks begin
29-30	~92%	~98%	Very low	Best performance	Optimum epoch
37	High	87.3%	Min (0.83)	Min val. loss	Optimal architecture
45+	Very high	Fluctuating	Slight rise	Overfitting	Accuracy may decline

VII. CONCLUSIONS

This project successfully demonstrated that a deep learning-based integrated framework can accurately classify forest fire and non-fire images while supporting real-time deployment. By combining CNNs and LSTMs, the system addressed challenges including limited labeled datasets, high visual variability, and the need for fast, stable predictions. The hybrid CNN–LSTM pipeline significantly improved feature extraction, temporal understanding, and operational reliability.

The CNN component captured fine spatial features including flame edges, smoke regions, vegetation patterns, and anomalous bright spots. The LSTM layers analyzed sequential patterns such as drifting smoke and flickering flames, reducing false alarms from static objects. Together they produced a reliable spatiotemporal model with accurate predictions across diverse forest types, seasons, and atmospheric conditions.

Optimization via pruning, quantization, and knowledge distillation enabled deployment on IoT devices, drones, and edge cameras. The model delivered strong accuracy, high precision, and reliable recall. Early detection minimizes ecological loss, reduces financial damage, and provides authorities with ample response time, demonstrating the potential of modern AI in transforming forest monitoring and management.

VIII. FUTURE SCOPE

Future directions include: (1) LiDAR integration for 3D forest modelling; (2) autonomous drone monitoring with edge-deployed CNN–LSTM; (3) IoT sensor network integration for improved early detection; (4) LSTM-based forecasting for fire spread prediction; (5) advanced augmentation for rare events; (6) GIS and government dashboard integration; (7) mobile application for field officers; (8) global dataset training for cross-region generalisation; (9) semi-supervised continual learning; and (10) automated forest health index for long-term vegetation monitoring.

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